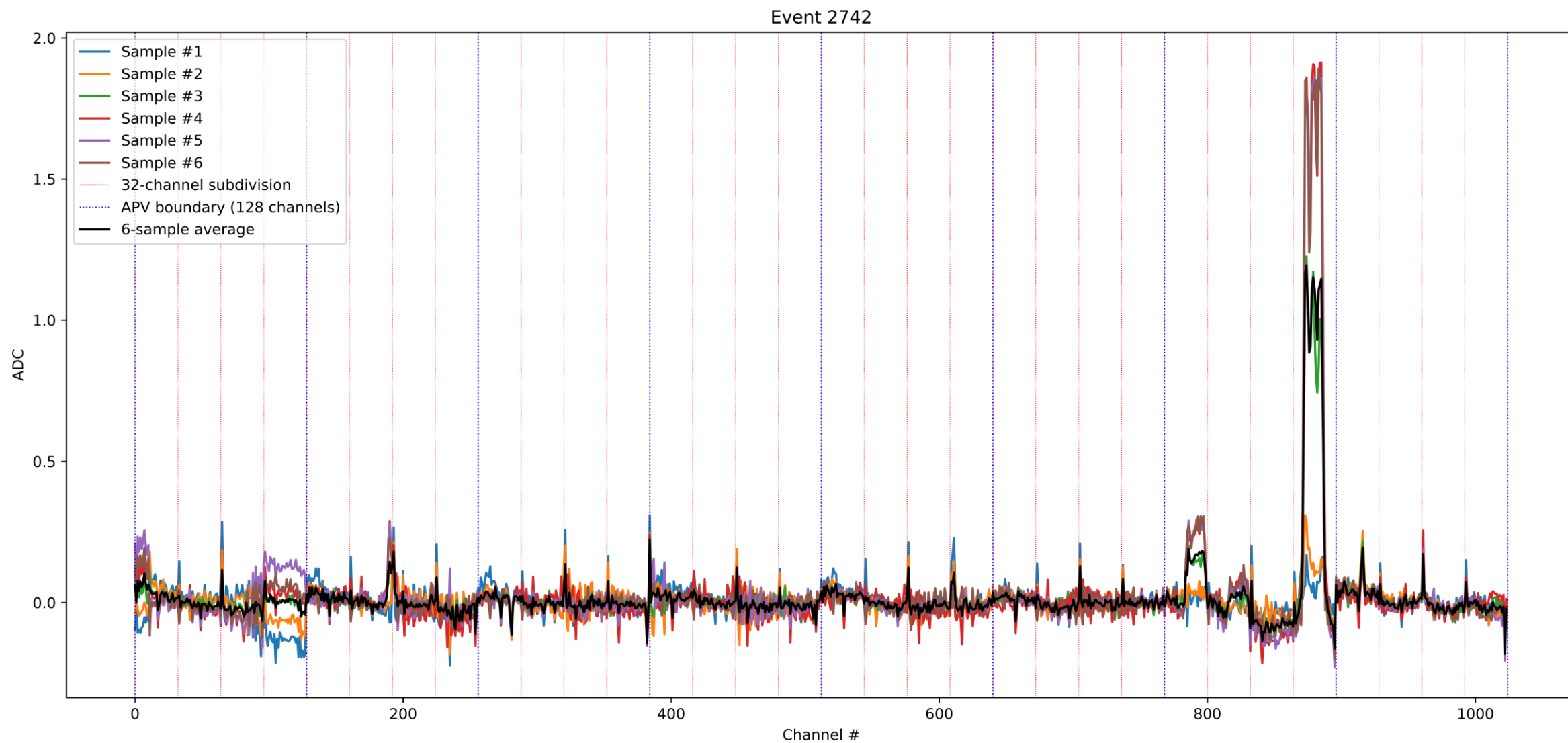


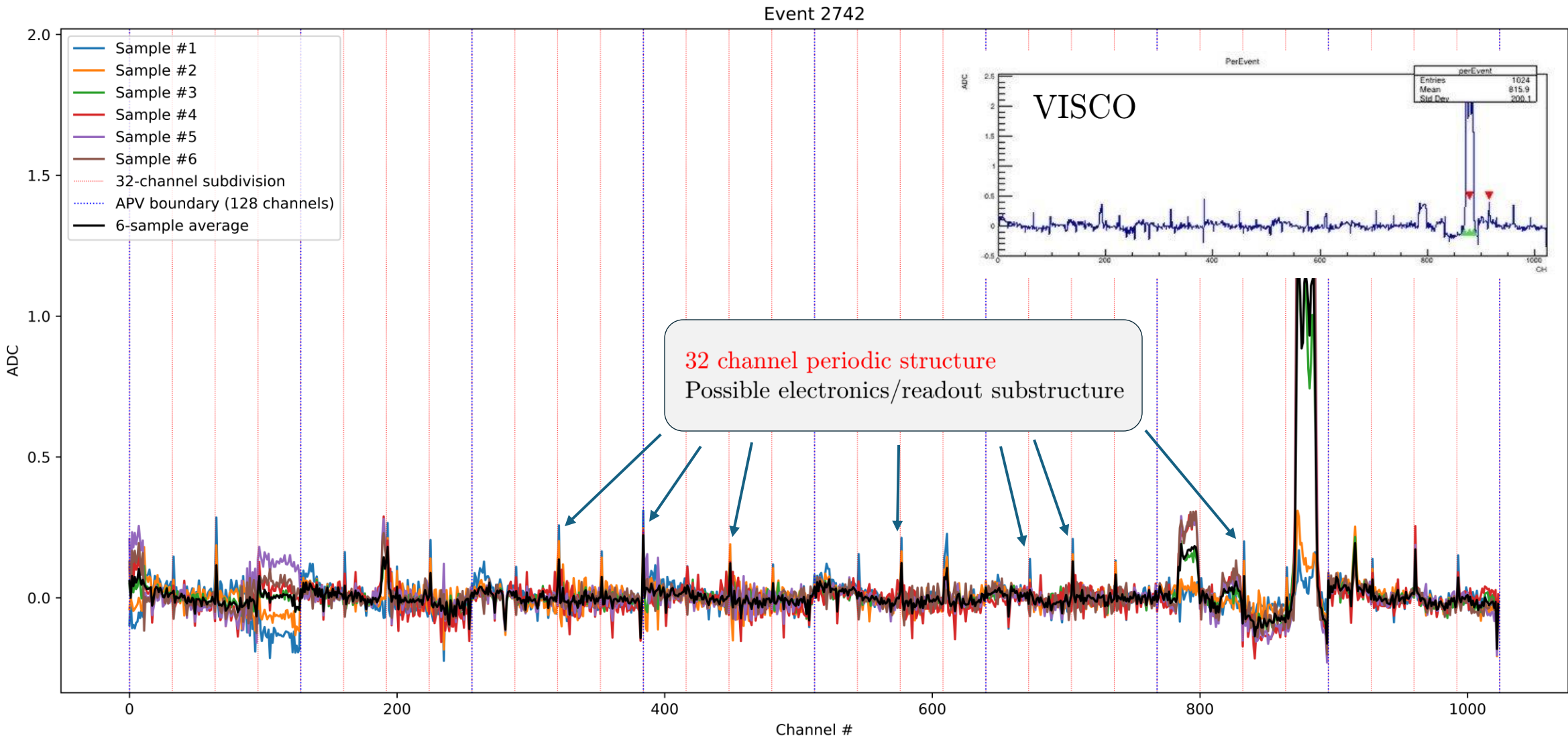
September 8, 2026

RGHF : Robust GEM Hit Finder

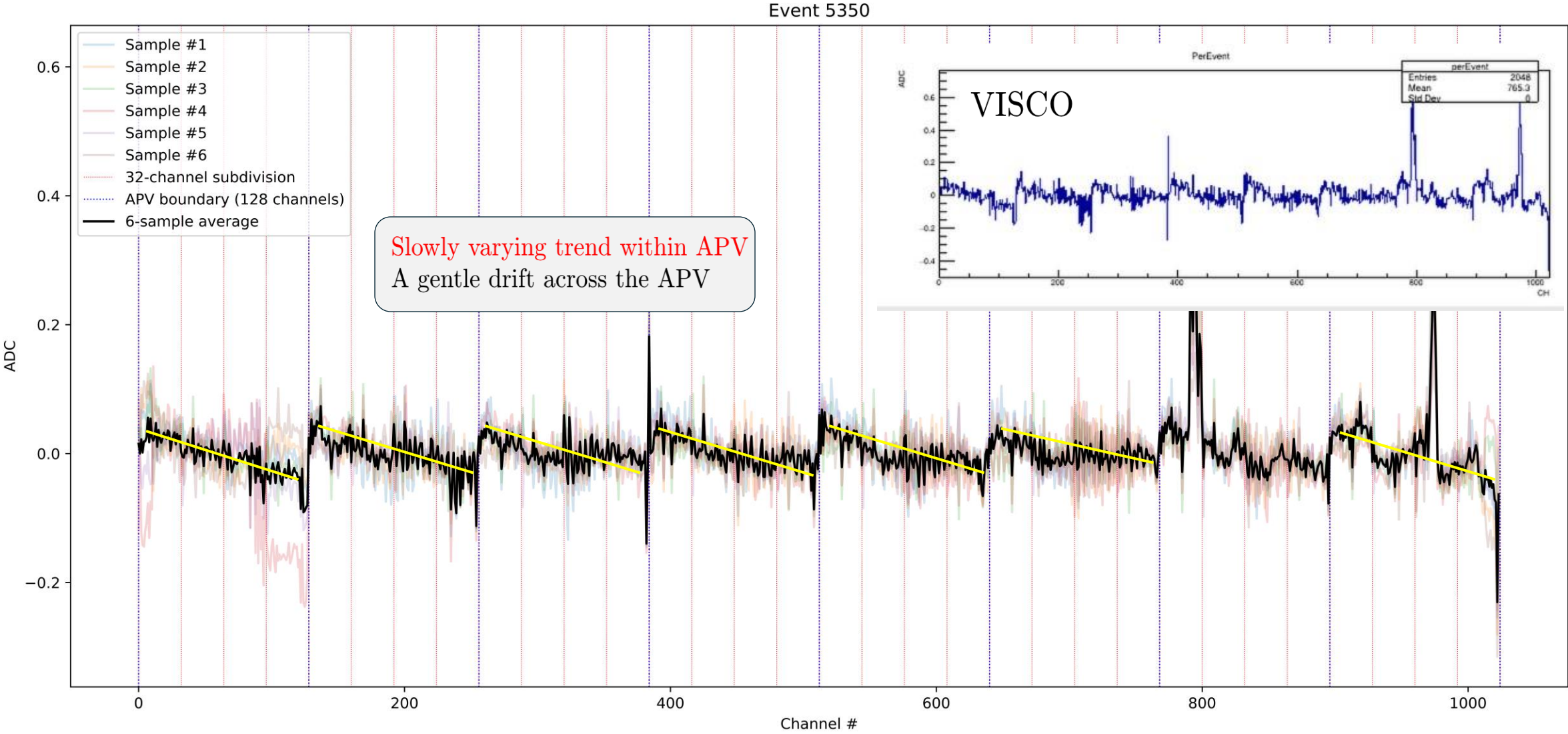
The Problem



The Problem

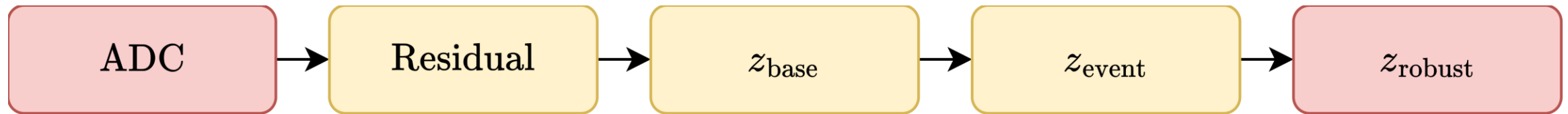


The Problem



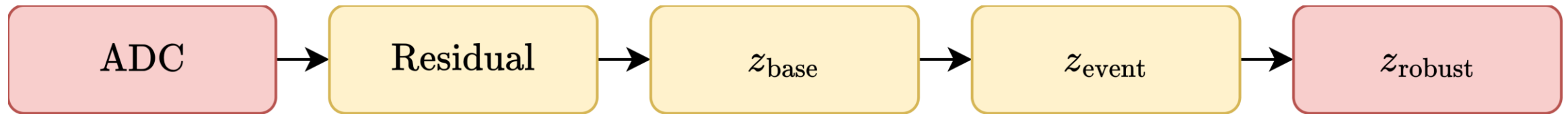
The Idea

I. Robust Noise Normalization

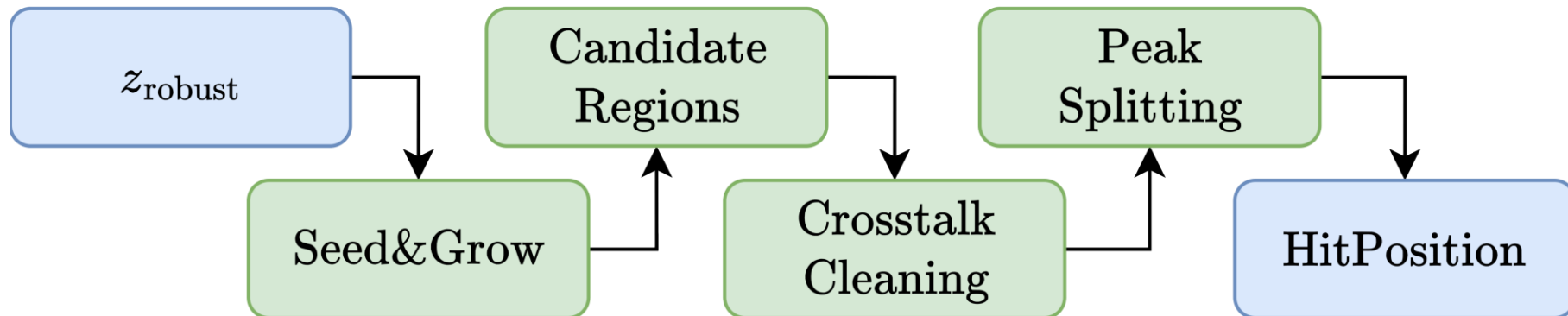


The Idea

I. Robust Noise Normalization



II. Hit Reconstruction



1. Residual

I. Robust Noise Normalization

- Choose a radius. We took radius = 15
- Let $A_{e,c}$ = ADC value corresponding to channel c for event e
- Make a median filter for each channel c :

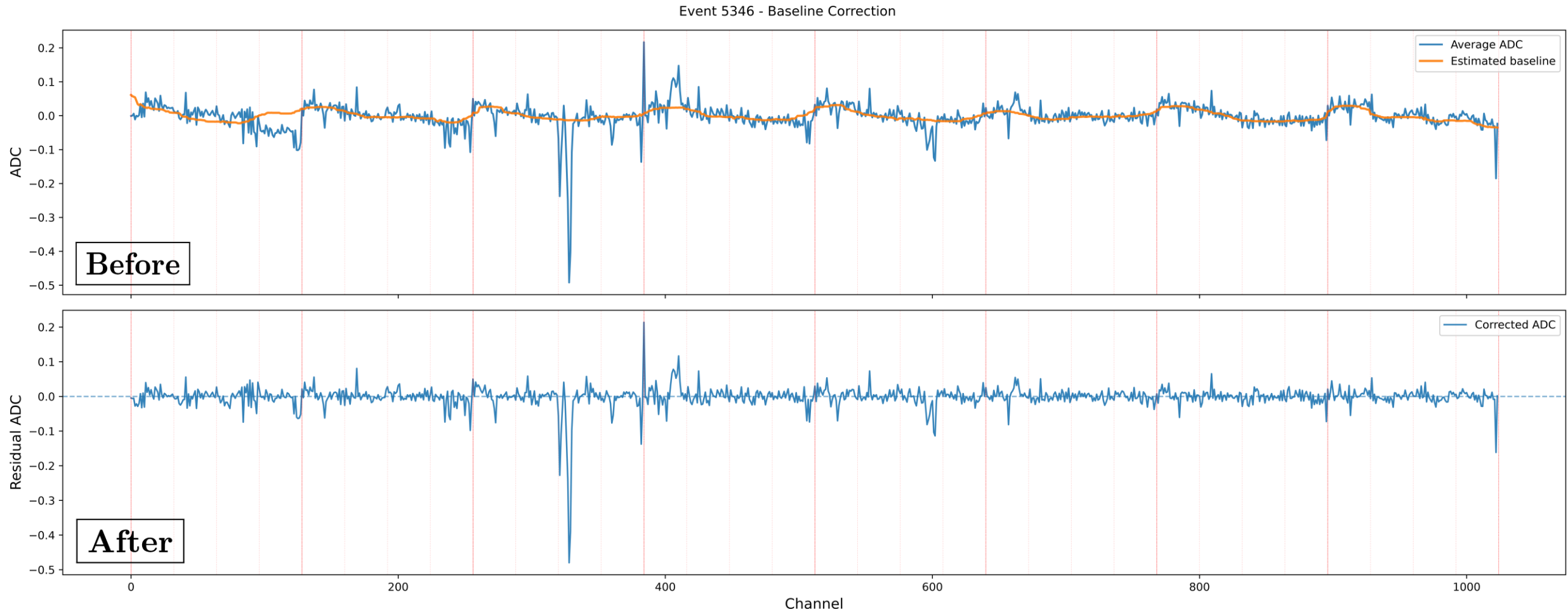
$$B_{e,c} = \text{median}\{A_{e,j} | j \in [c - 15, c + 15] \cap [0, N - 1]\}$$

- This gives residual^s (or trend corrected ADC):

$$r_{e,c} = A_{e,c} - B_{e,c} \quad c = 1, \dots, 1024$$

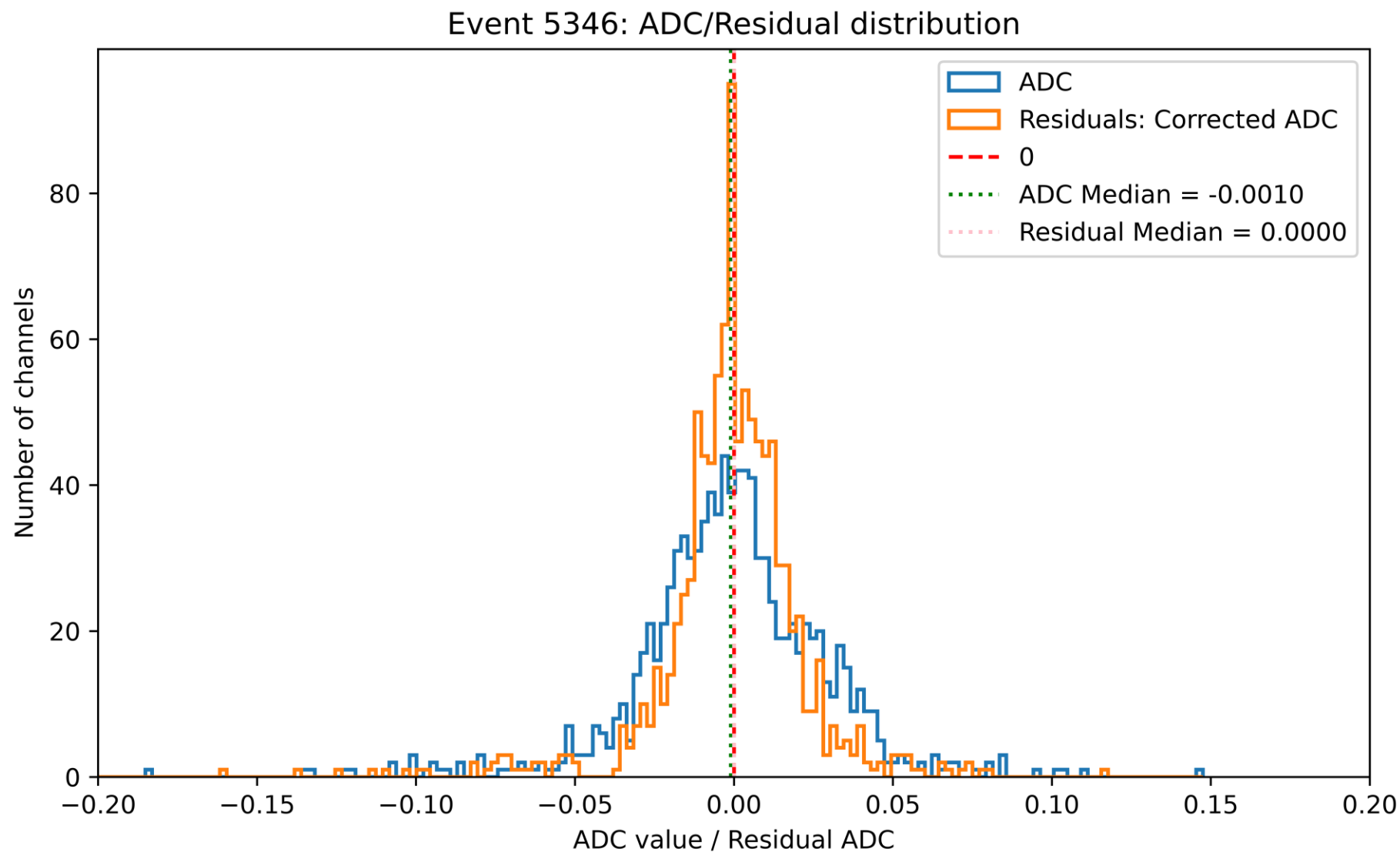
1. Residual

I. Robust Noise Normalization



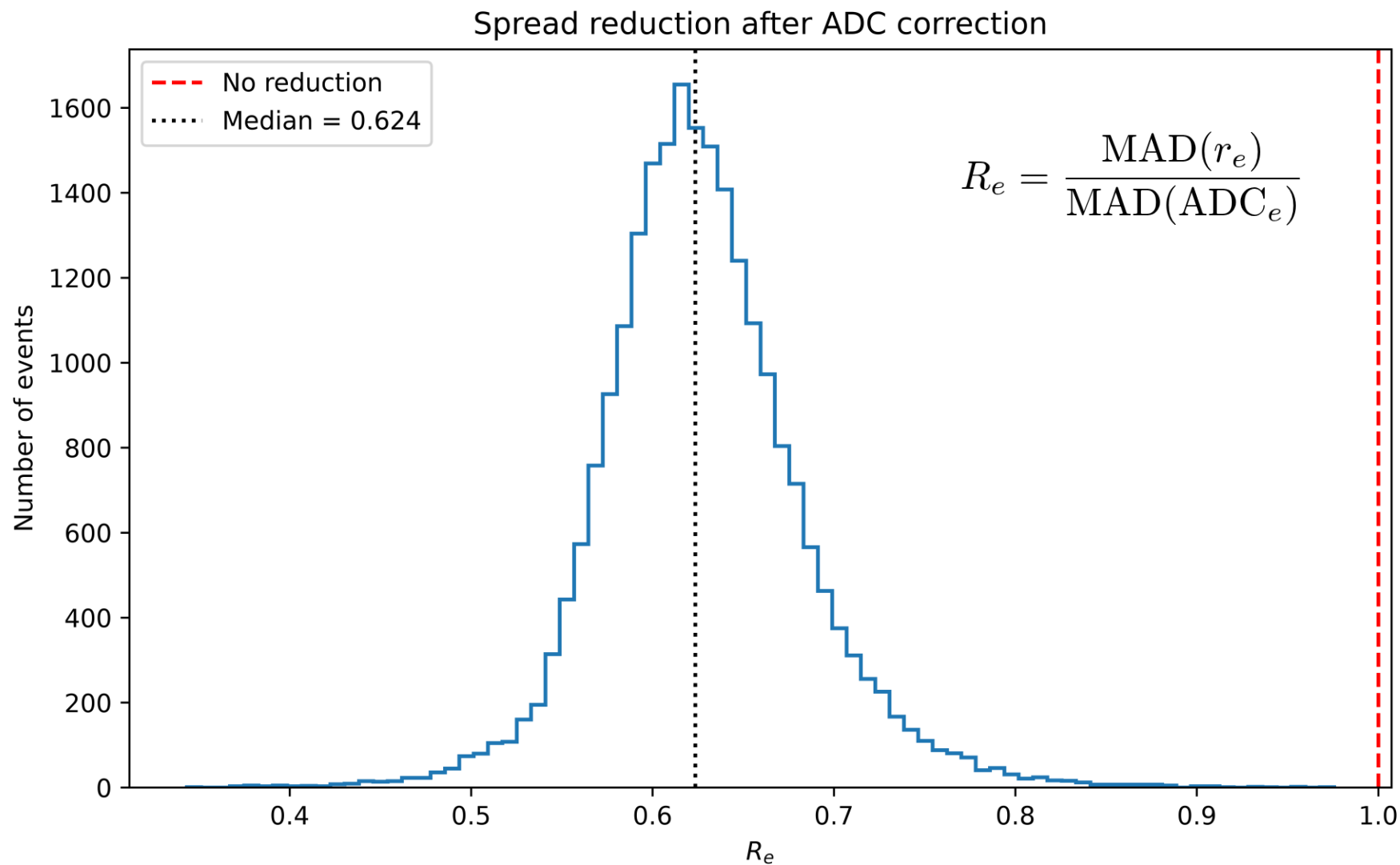
1. Residual

I. Robust Noise Normalization



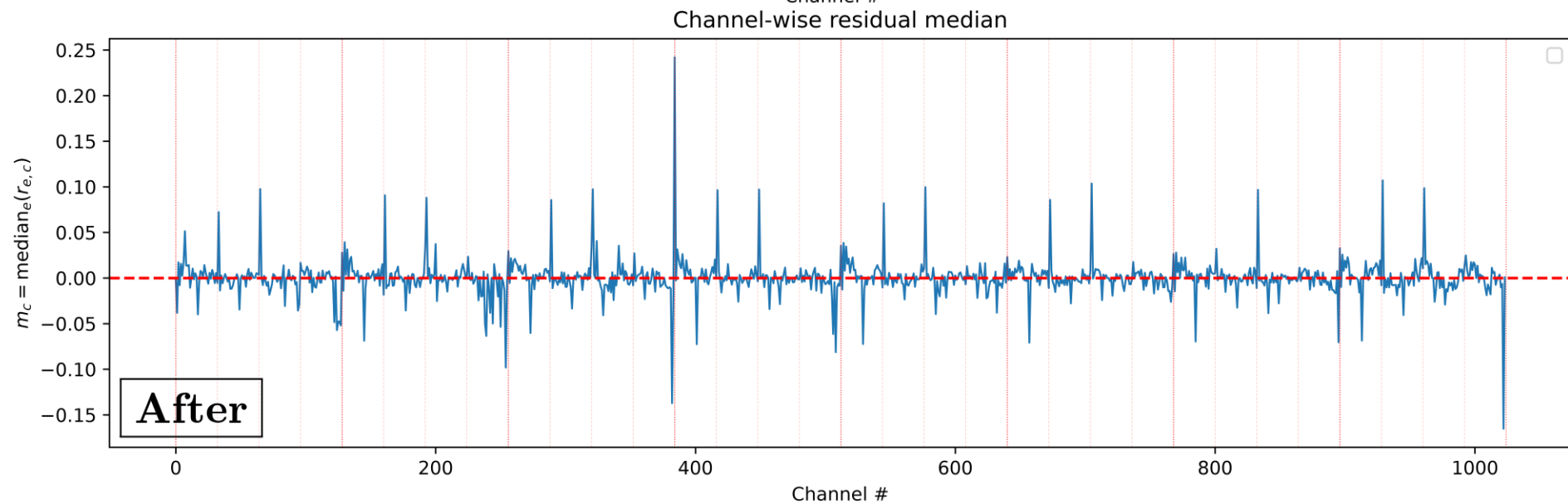
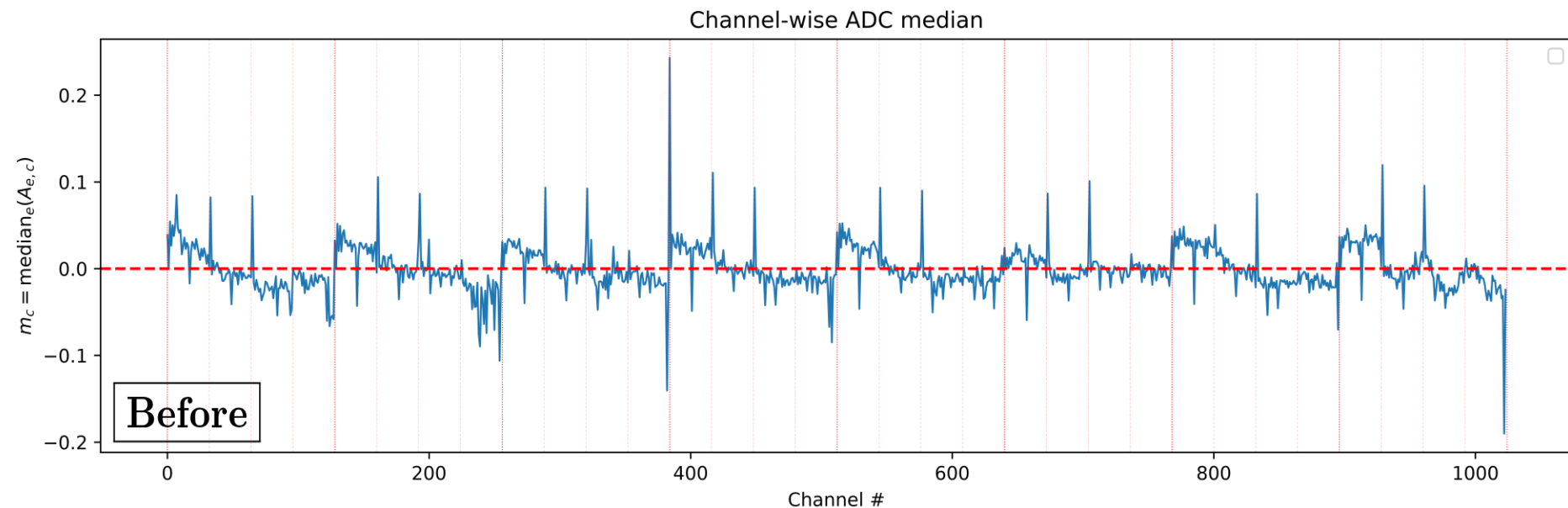
1. Residual

I. Robust Noise Normalization



1. Residual

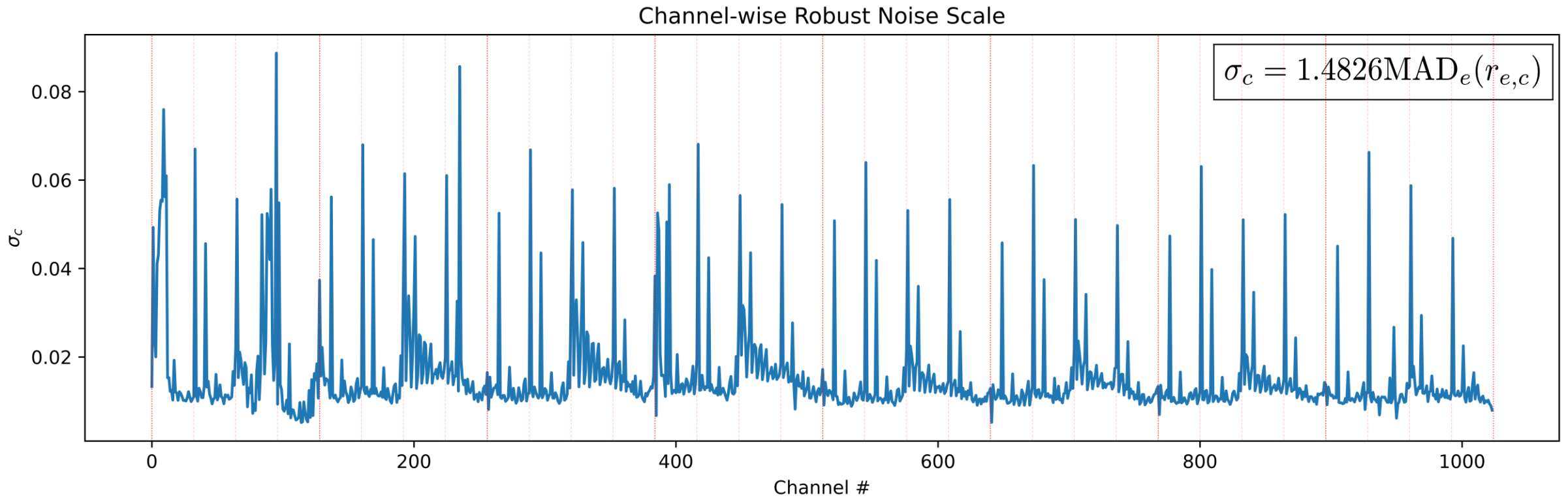
I. Robust Noise Normalization



1. Residual

I. Robust Noise Normalization

- $\text{MAD}_e(r_{e,c}) = \text{median}_e |r_e - m_e|$ where $m_c = \text{median}_e(r_{e,c})$



2. z_{base} : Normalization

I. Robust Noise Normalization

- Normalize using:

$$m_c = \text{median}_e(r_{e,c}), \quad \text{MAD}_e = \text{median}_e(|r_{e,c} - m_c|)$$

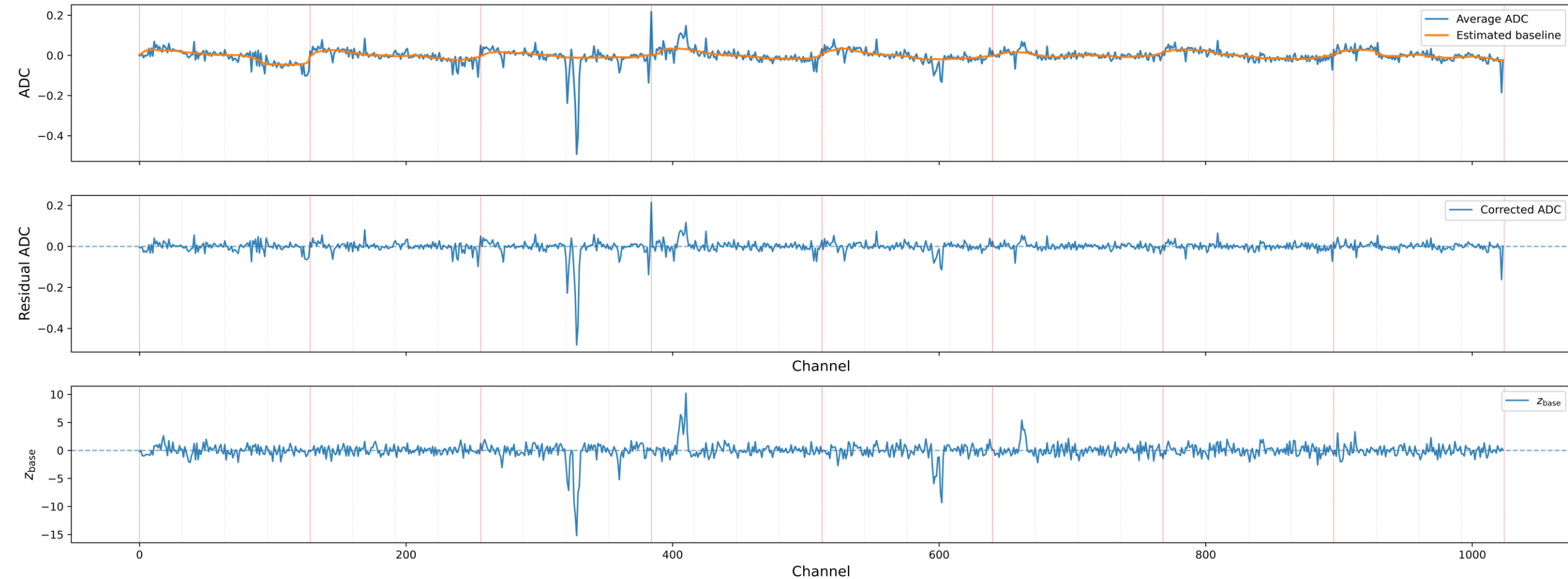
- Gaussian calibration factor is 1.4286, hence $\sigma^{\text{robust}} \approx 1.4826 \times \text{MAD}_e$
- The z-score would thus be:

$$z_{e,c} = \frac{r_{e,c} - m_c}{\sigma_c^{\text{robust}}}$$

2. z_{base} : Normalization

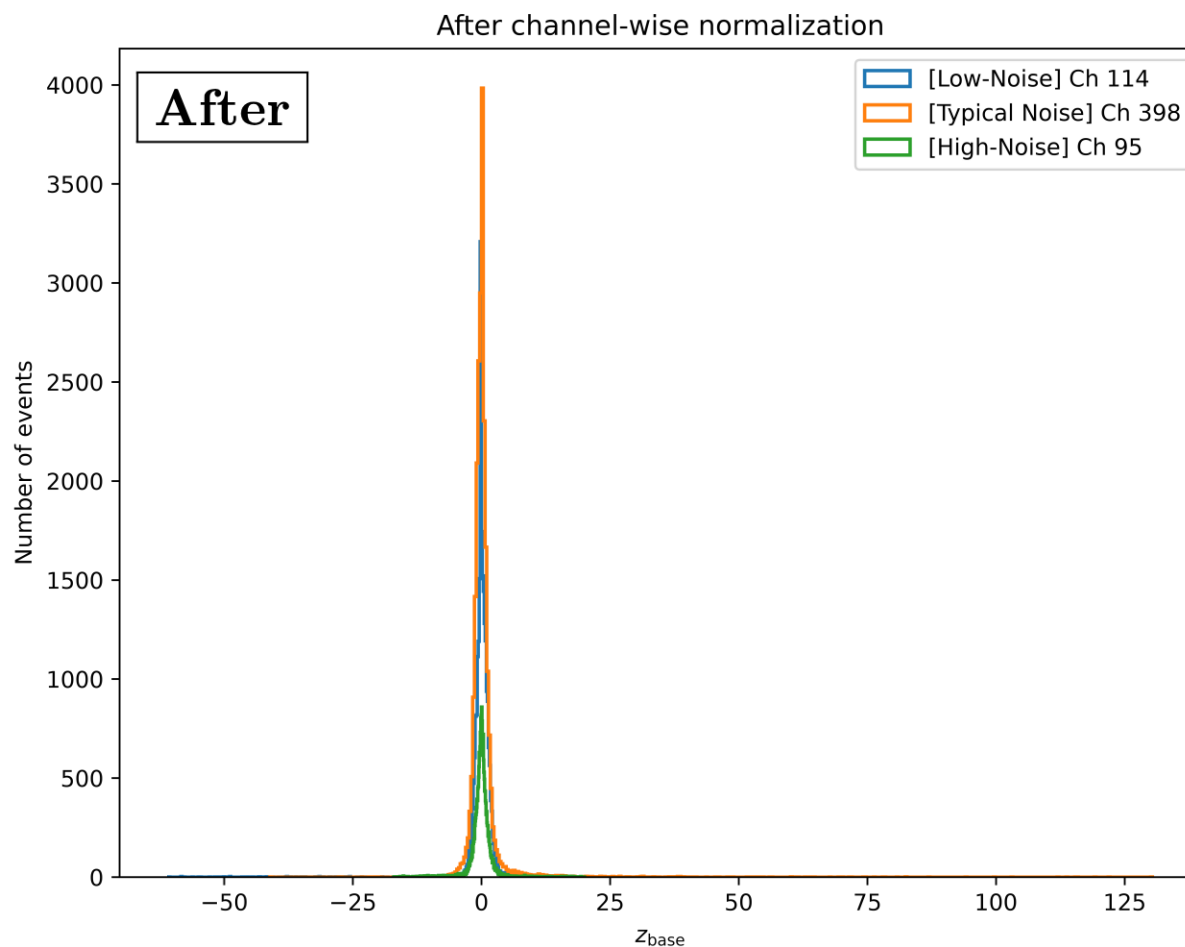
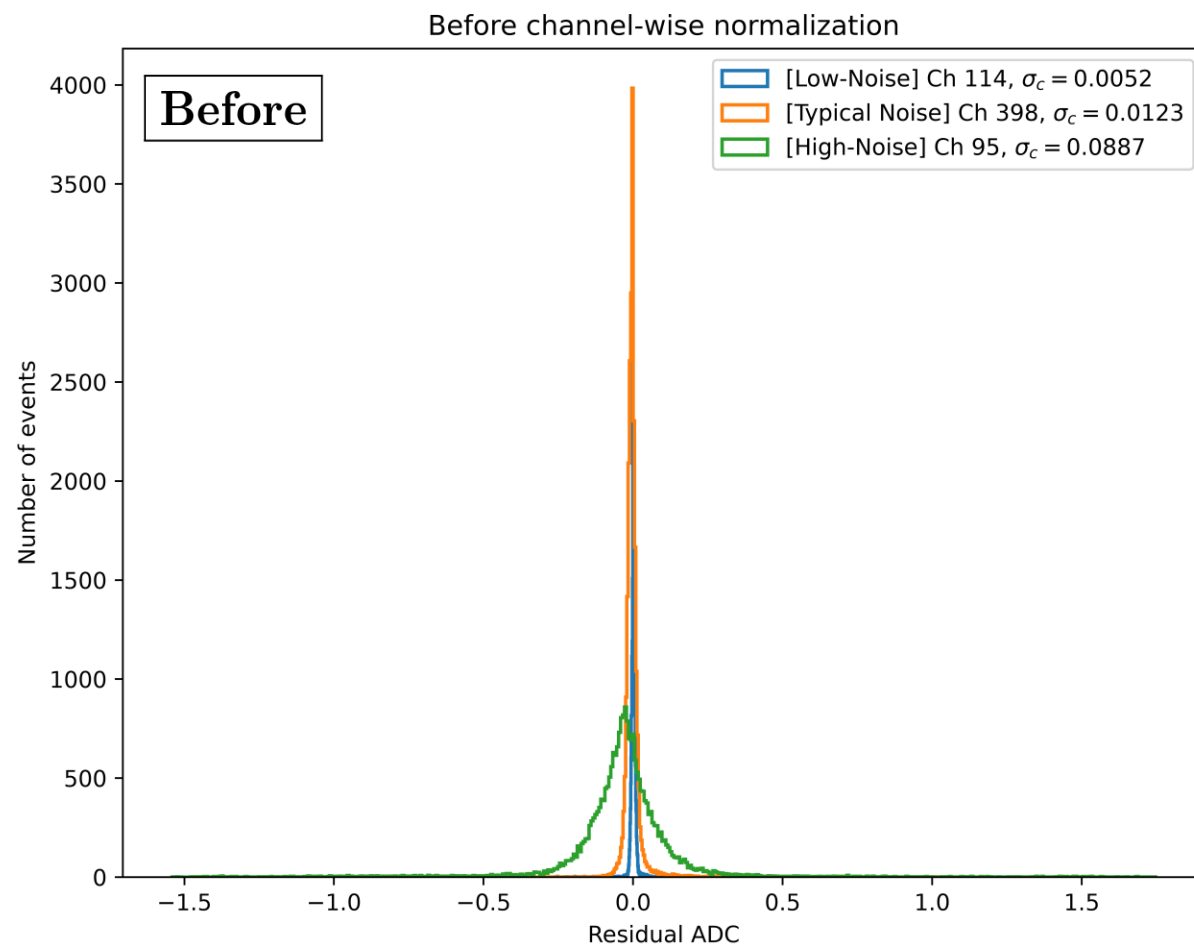
I. Robust Noise Normalization

Event 5346



2. z_{base} : Normalization

I. Robust Noise Normalization



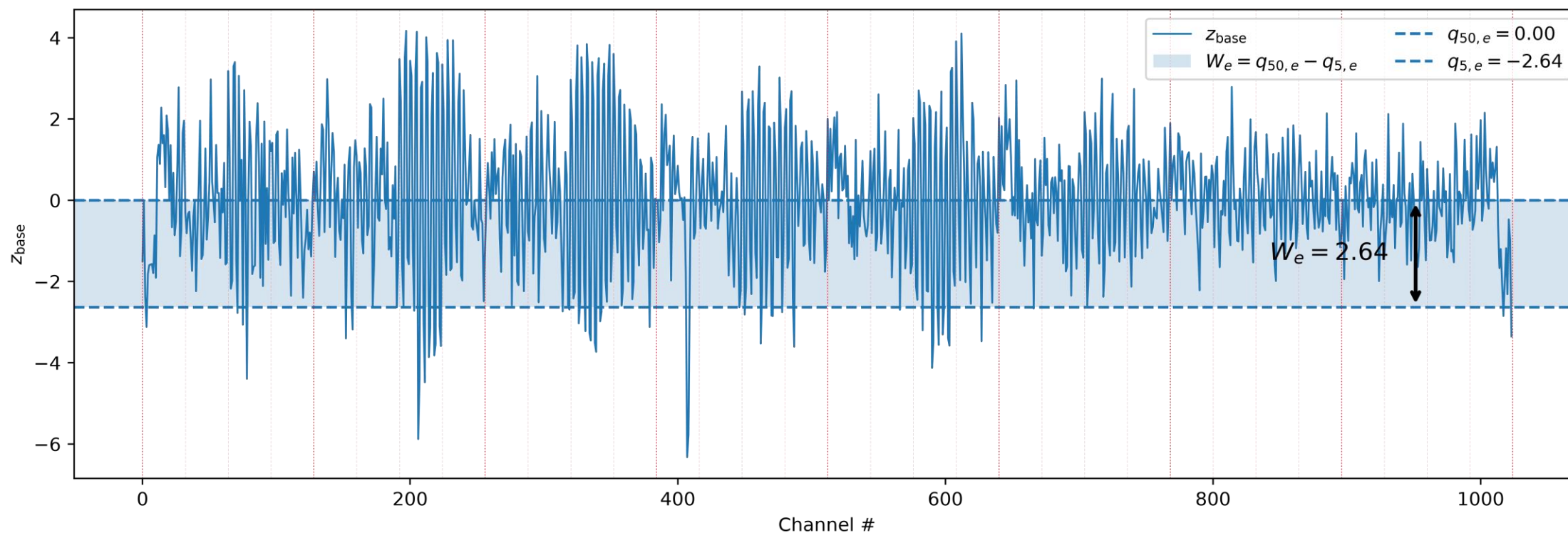
3. z_{event} : Event-Wise Noise I. Robust Noise Normalization

THE IDEA: Estimate the noise from the side of the distribution where you don't expect to see a signal!

- We first quantify the negative tail of the z dist. in the valid channels i.e. the region of in z -space where the z -values are lower than the typical value (median value) for that event. This is quantified using:

$$W_e = q_{50,e} - q_{05,e}$$

Event 12: Event-wide lower-tail noise estimate

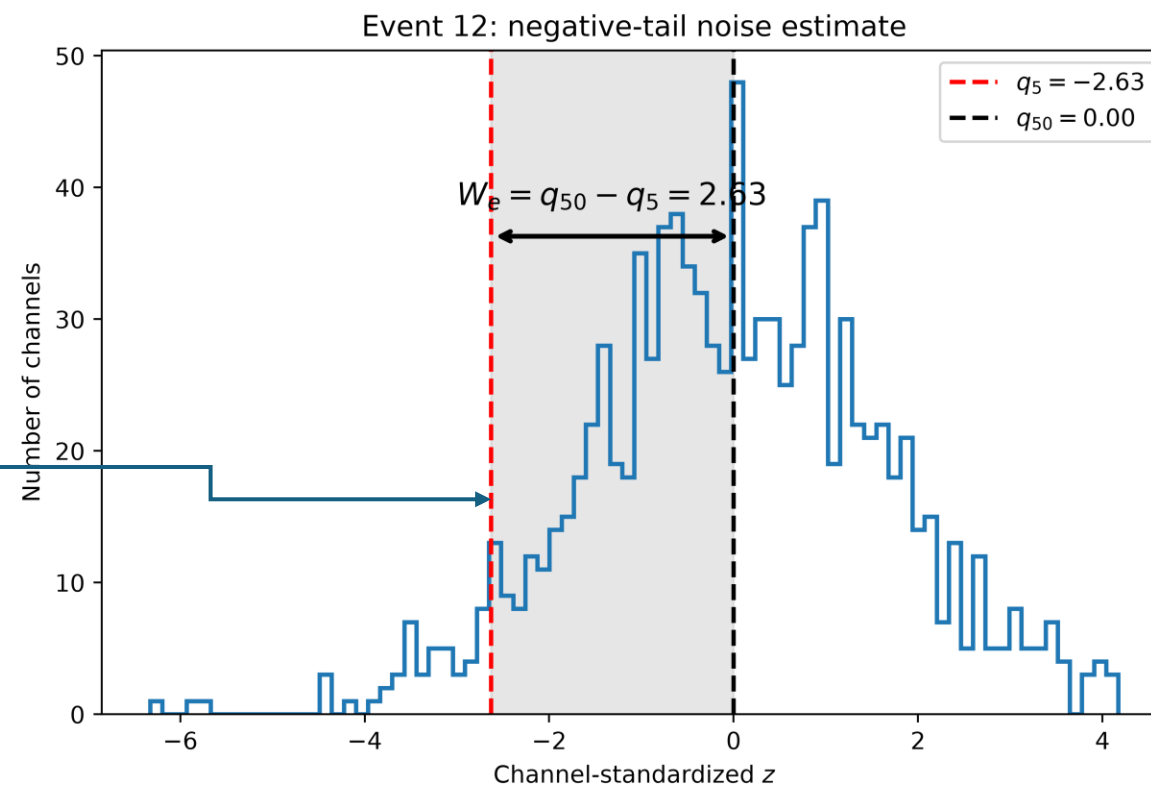
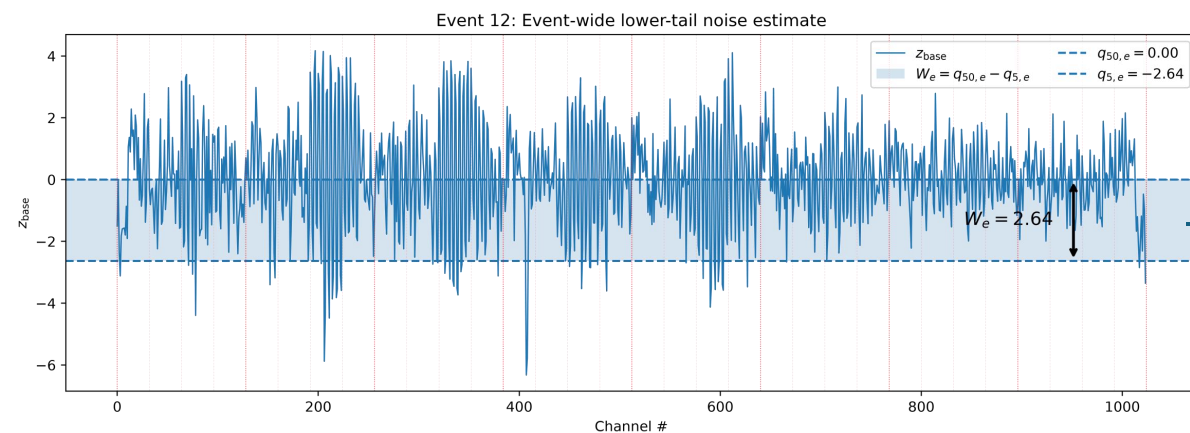


3. z_{event} : Event-Wise Noise I. Robust Noise Normalization

THE IDEA: Estimate the noise from the side of the distribution where you don't expect to see a signal!

- We first quantify the negative tail of the z dist. in the valid channels i.e. the region of in z -space where the z -values are lower than the typical value (median value) for that event. This is quantified using:

$$W_e = q_{50,e} - q_{05,e}$$

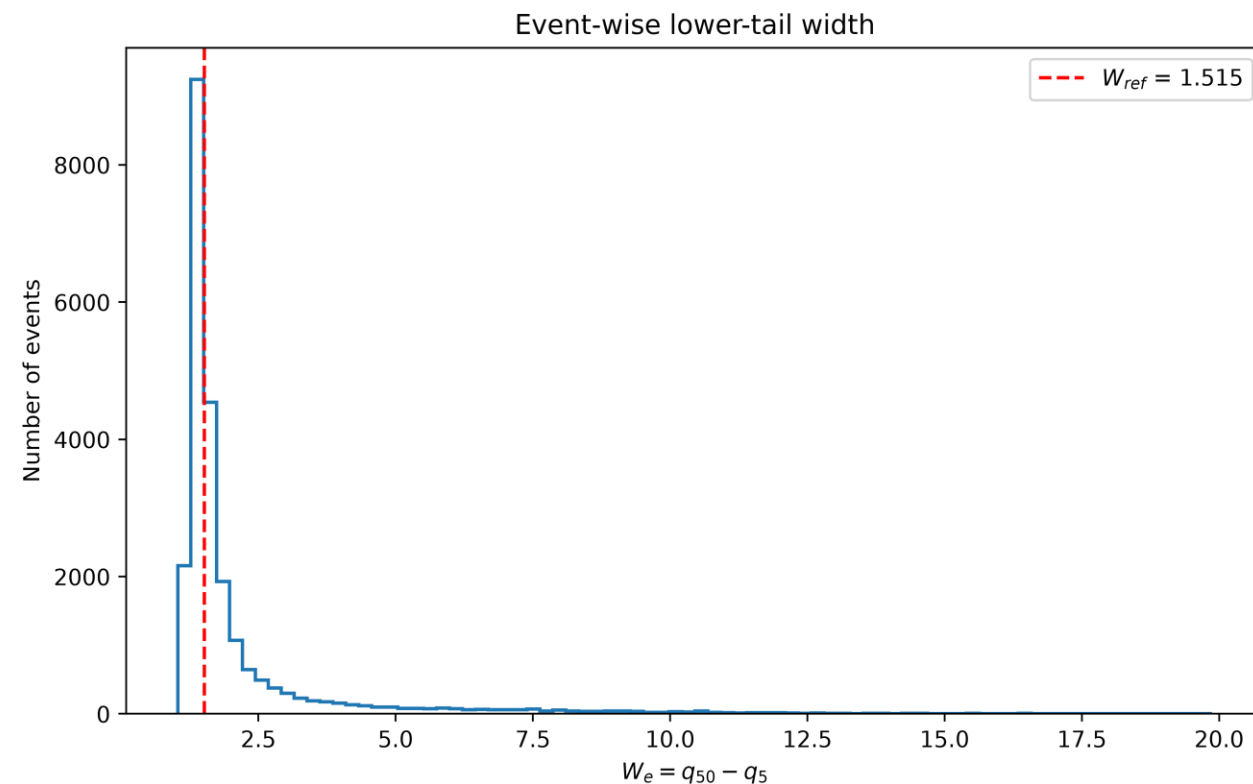
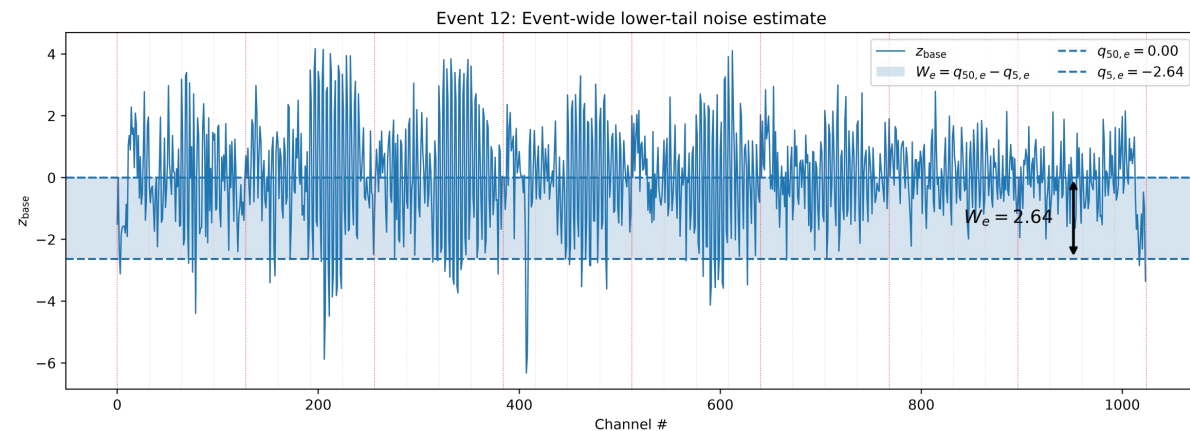


3. z_{event} : Event-Wise Noise I. Robust Noise Normalization

THE IDEA: Estimate the noise from the side of the distribution where you don't expect to see a signal!

- Then, we have a similar estimate for a typical event in the dataset. We use it for reference.

$$W_{\text{ref}} = \text{median}_e(W_e)$$

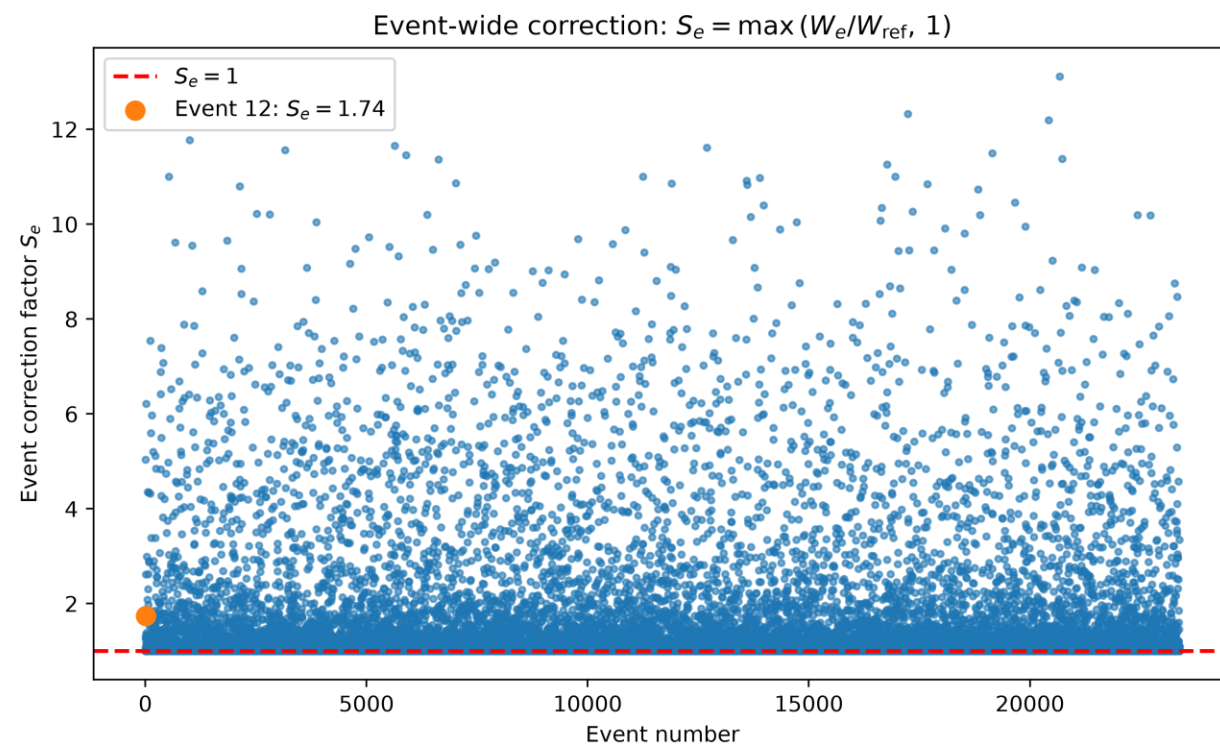
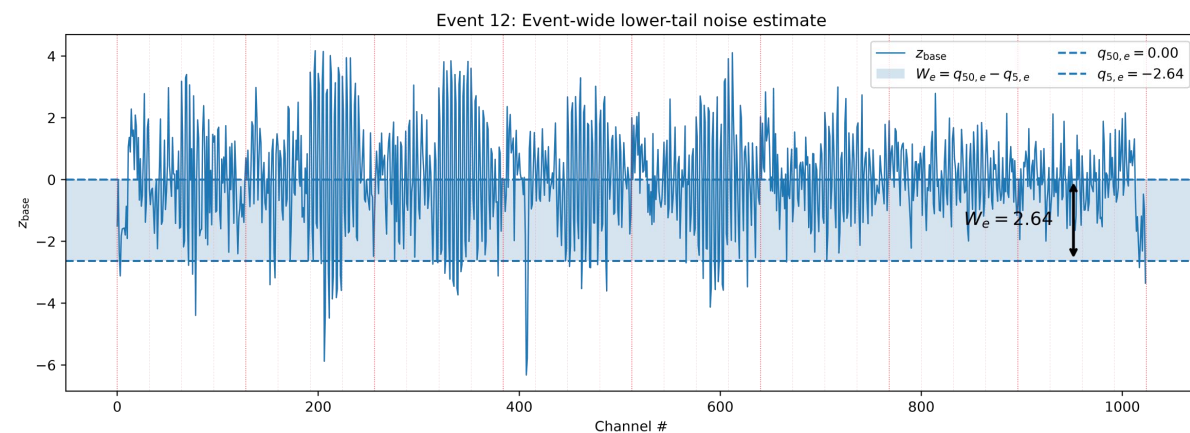


3. z_{event} : Event-Wise Noise I. Robust Noise Normalization

THE IDEA: Estimate the noise from the side of the distribution where you don't expect to see a signal!

- Event noise scale $S_e \geq 1$ given by

$$S_e = \max\left\{\frac{W_e}{W_{\text{ref}}}, 1\right\}$$

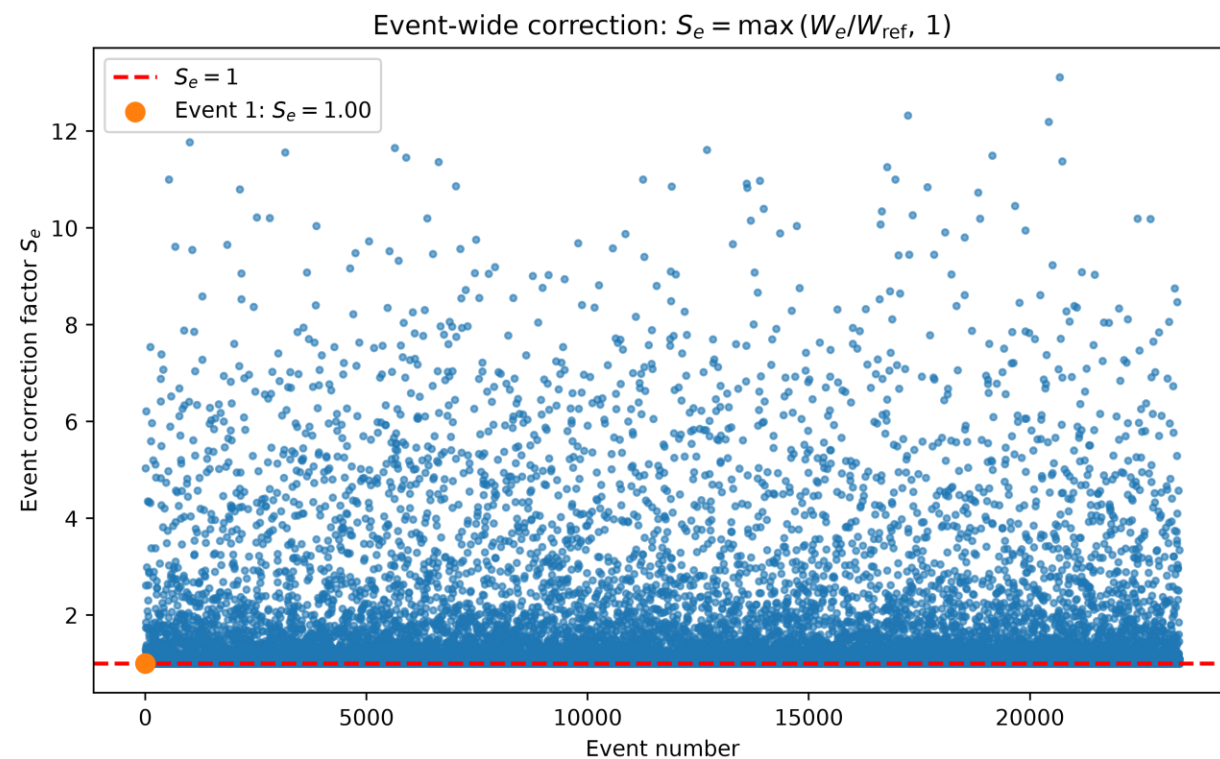
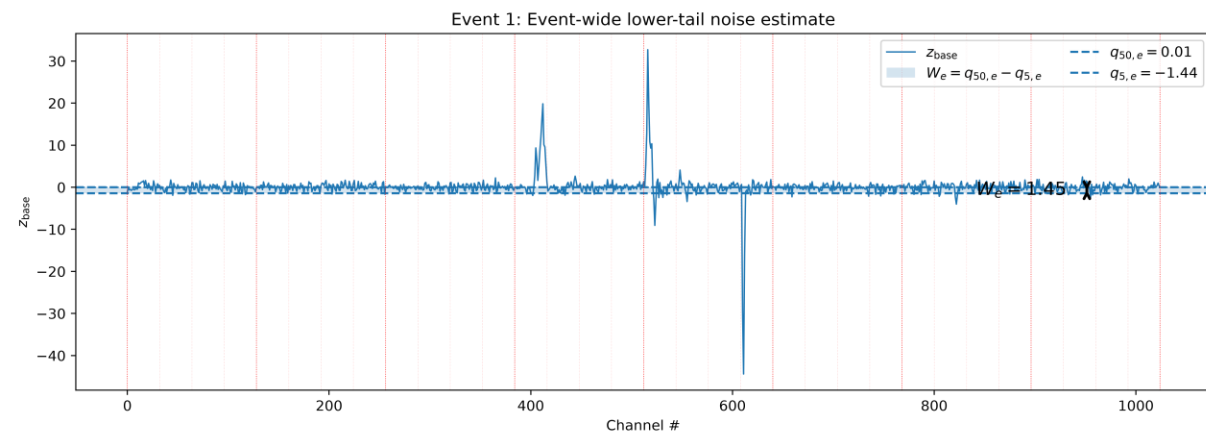


3. z_{event} : Event-Wise Noise I. Robust Noise Normalization

THE IDEA: Estimate the noise from the side of the distribution where you don't expect to see a signal!

- Event noise scale $S_e \geq 1$ given by

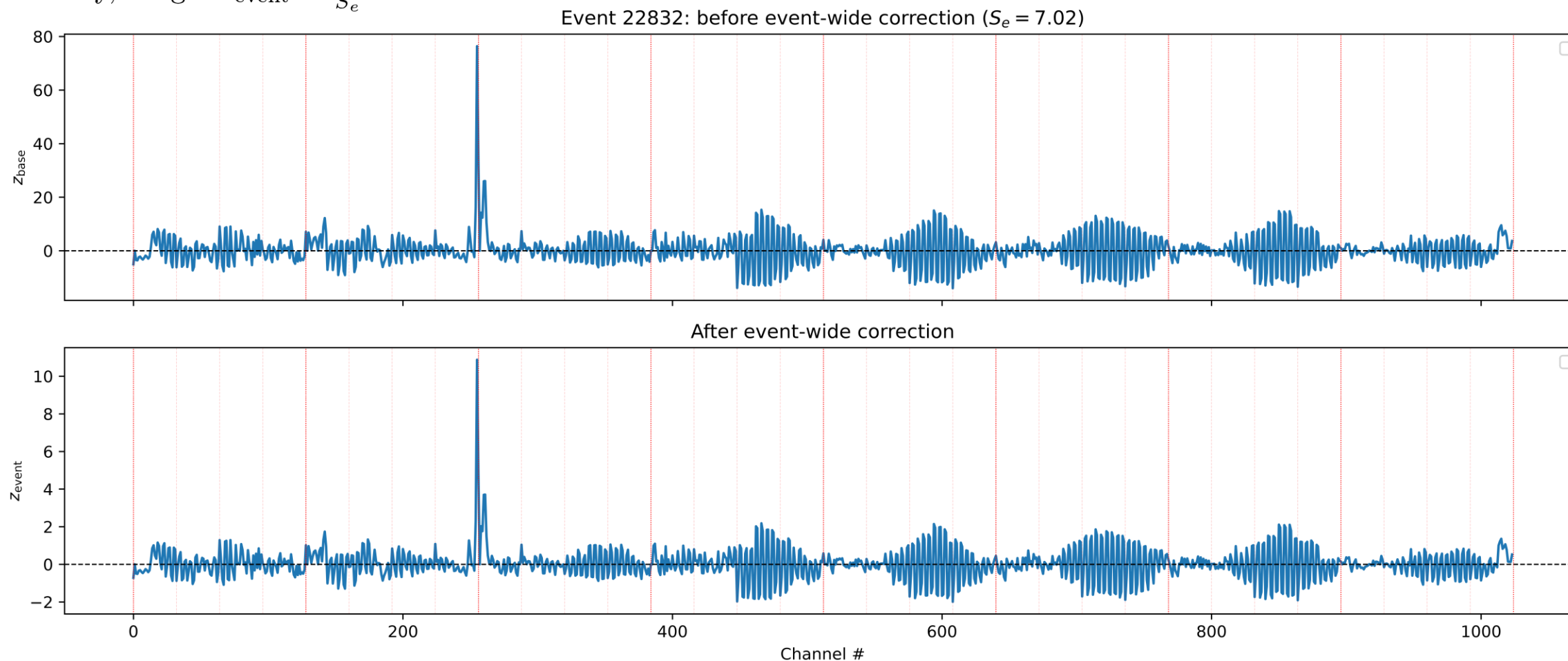
$$S_e = \max\left\{\frac{W_e}{W_{\text{ref}}}, 1\right\}$$



3. z_{event} : Event-Wise Noise I. Robust Noise Normalization

THE IDEA: Estimate the noise from the side of the distribution where you don't expect to see a signal!

- Finally, we get $z_{\text{event}} = \frac{z}{S_e}$



4. z_{robust} : APV-Wide Noise I. Robust Noise Normalization

- Calculate APV-specific scale from the original z :

$$S_{e,a}^{\text{APV}} = \max \left\{ \frac{W_{e,c}}{W_{\text{ref},a}}, 1 \right\} \quad a \in \{1, 2, 3, 4, 5, 6, 7, 8\}$$

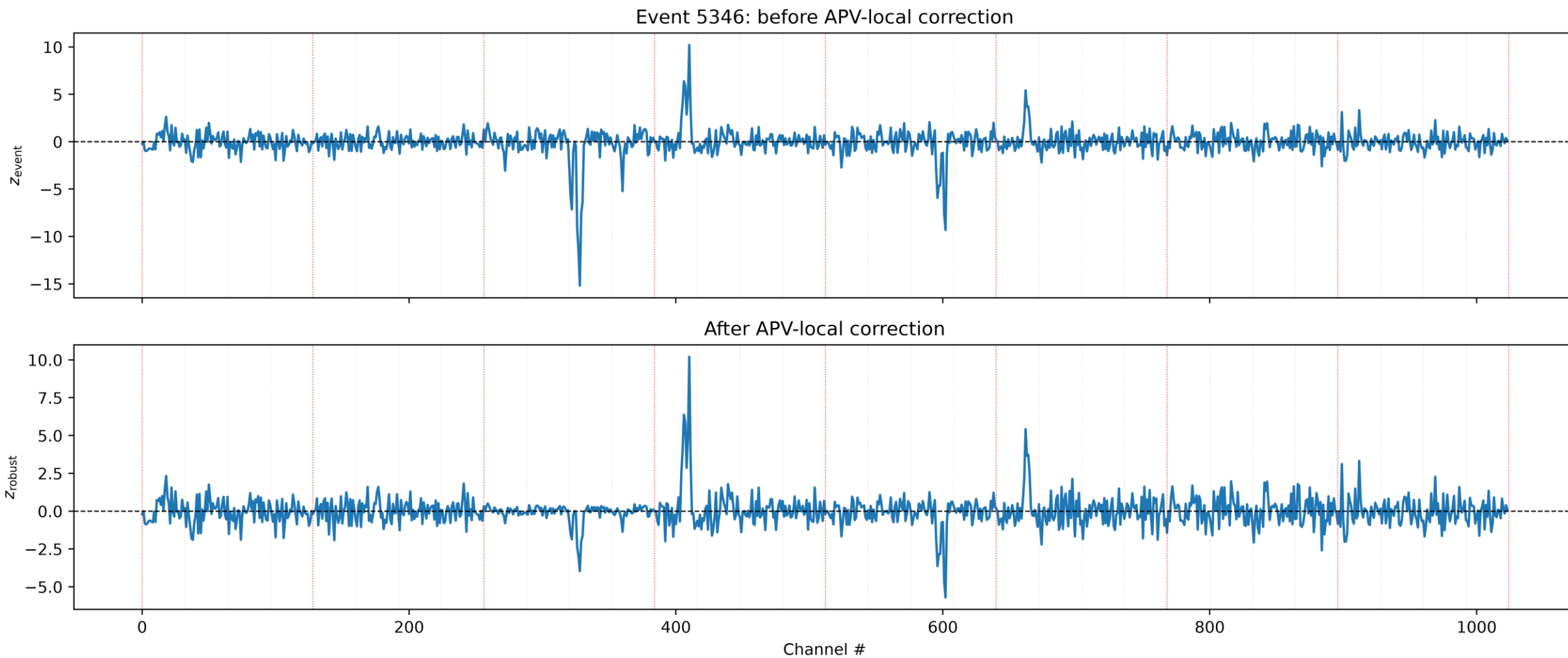
- We then take care to avoid double correction:

$$S_{e,a}^{\text{extra}} = \max \left\{ \frac{S_{e,a}^{\text{APV}}}{S_e}, 1 \right\}$$

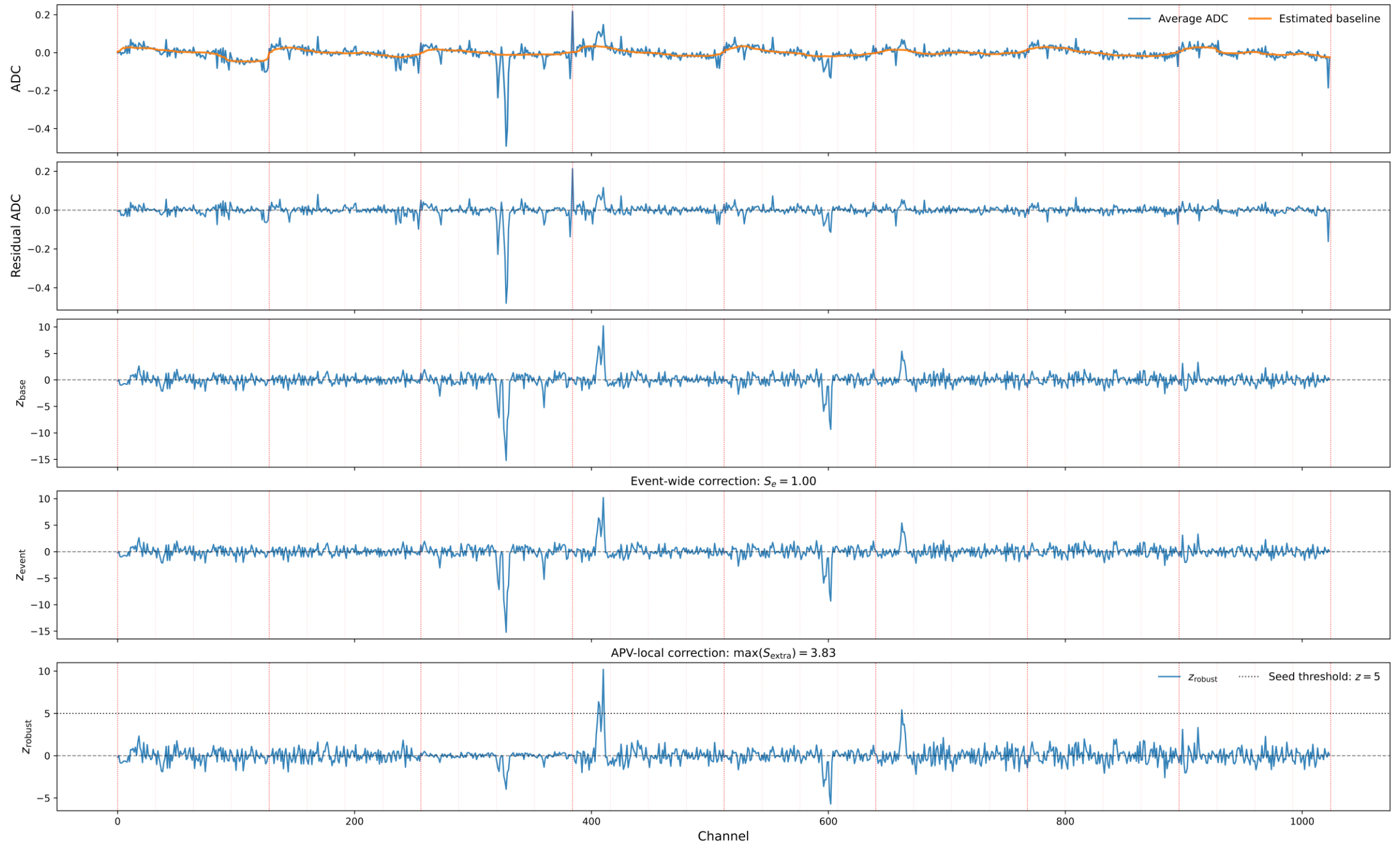
- Using the correction, we build our robust z-score:

$$z_{e,c}^{\text{robust}} = \frac{z_{e,c}}{S_e \times S_{e,a(c)}^{\text{extra}}}$$

4. z_{robust} : APV-Wide Noise I. Robust Noise Normalization



Event 5346: ADC → Robustly Normalized Signal



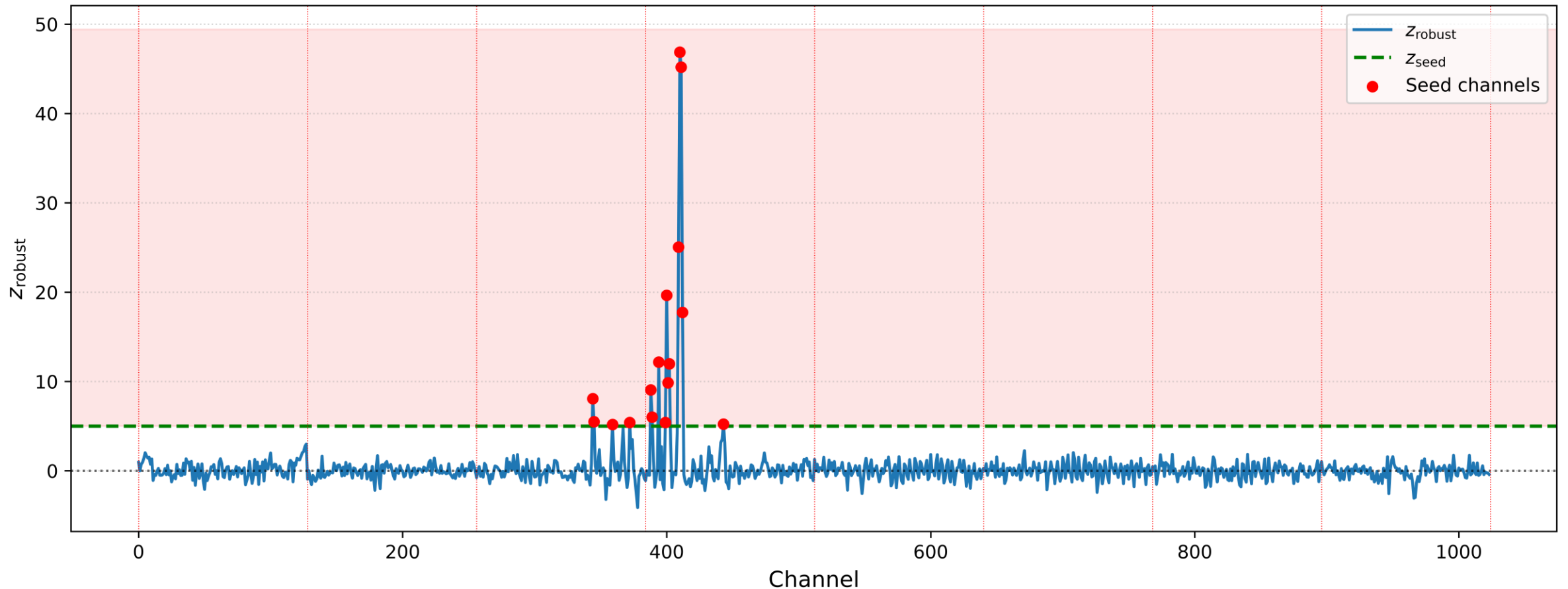
1. Seed&Grow

II. Hit Reconstruction

- A channel is selected as a **seed** if

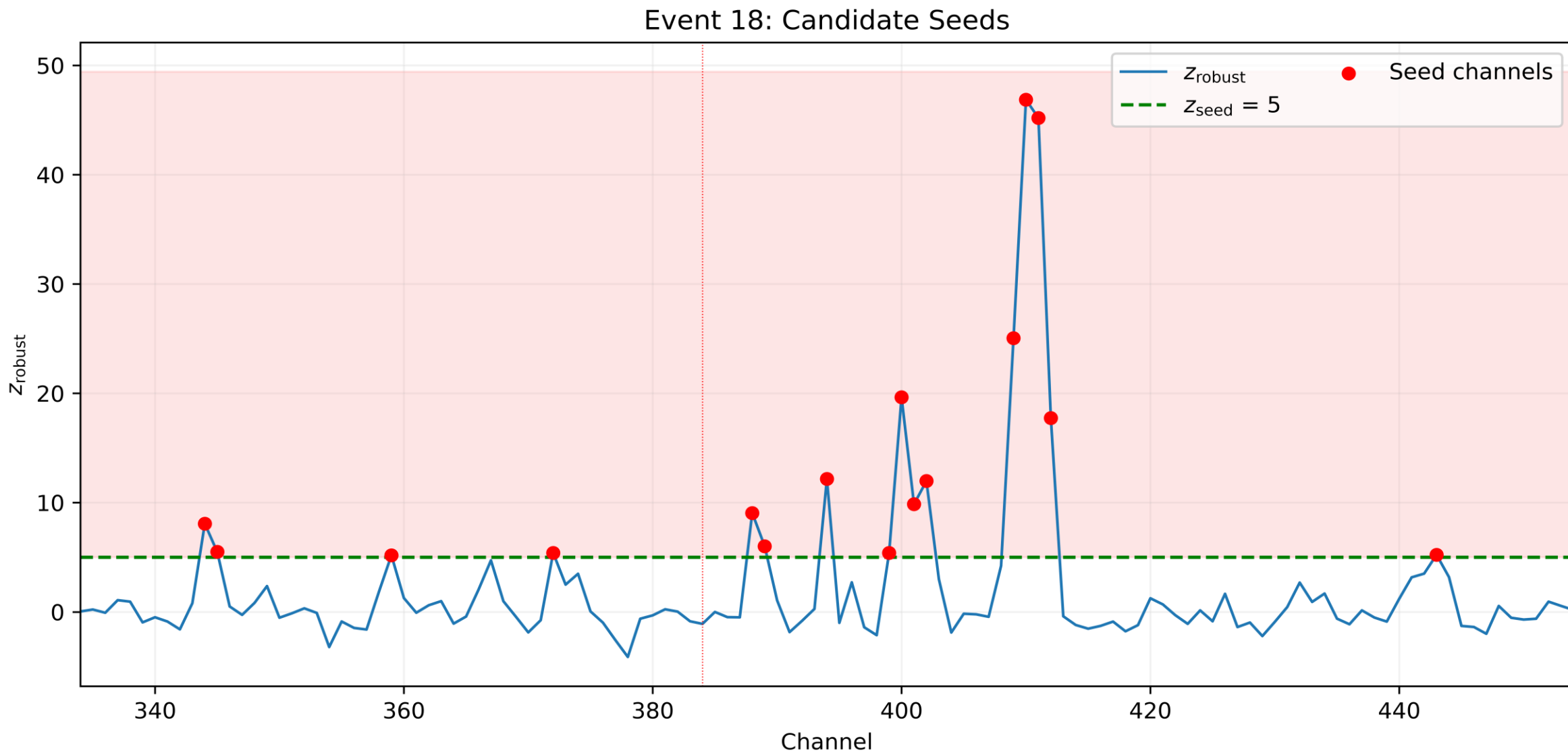
$$z_{\text{robust}}(c) > z_{\text{seed}},$$

Event 18: Candidate Seeds



1. Seed&Grow

II. Hit Reconstruction



1. Seed&Grow

II. Hit Reconstruction

- A channel is selected as a **seed** if

$$z_{\text{robust}}(c) > z_{\text{seed}},$$

- Starting from each seed, the candidate cluster is expanded independently to the left and right while neighboring channels satisfy

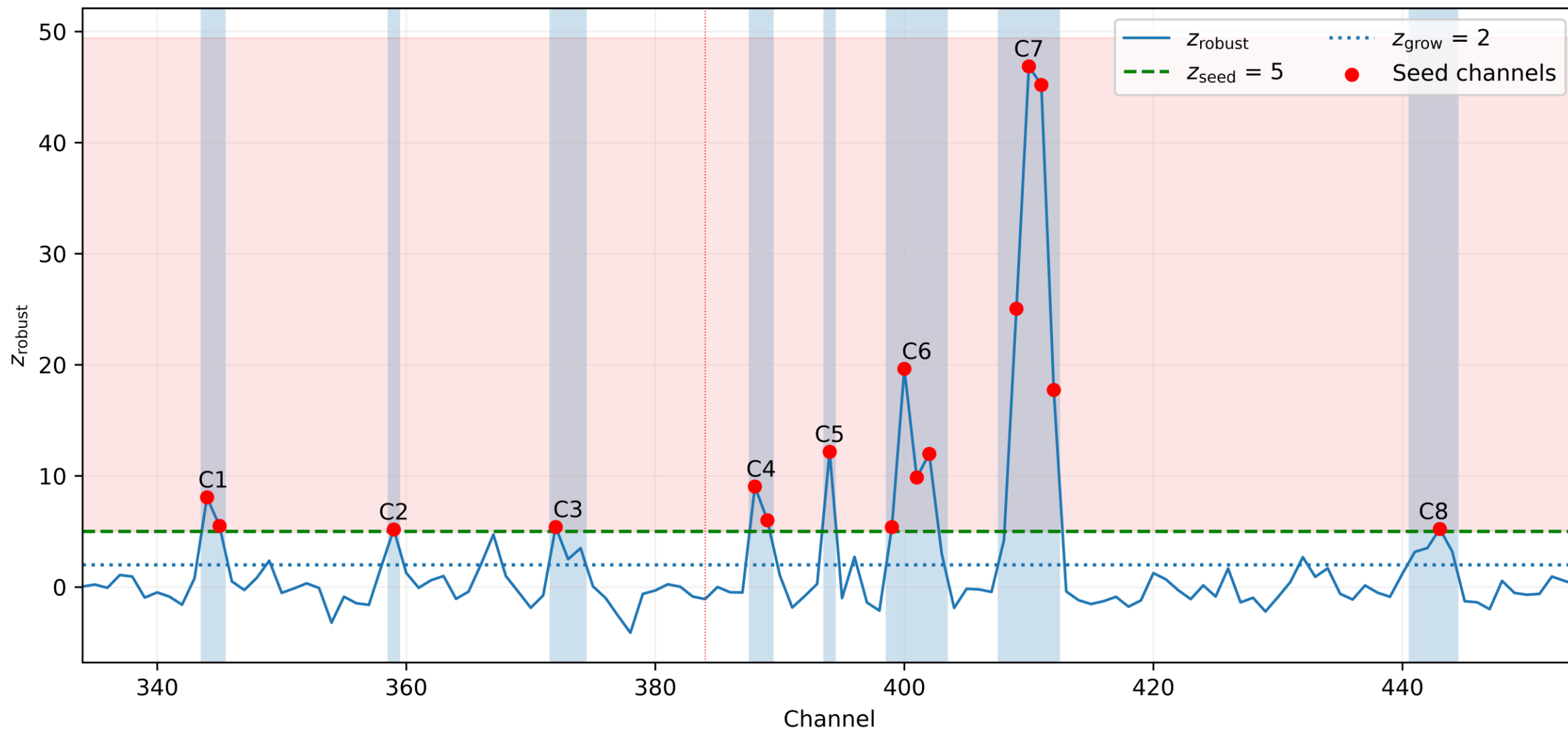
$$z_{\text{robust}}(c) > z_{\text{grow}},$$

- Growth stops when the signal falls below the grow threshold.

1. Seed&Grow

II. Hit Reconstruction

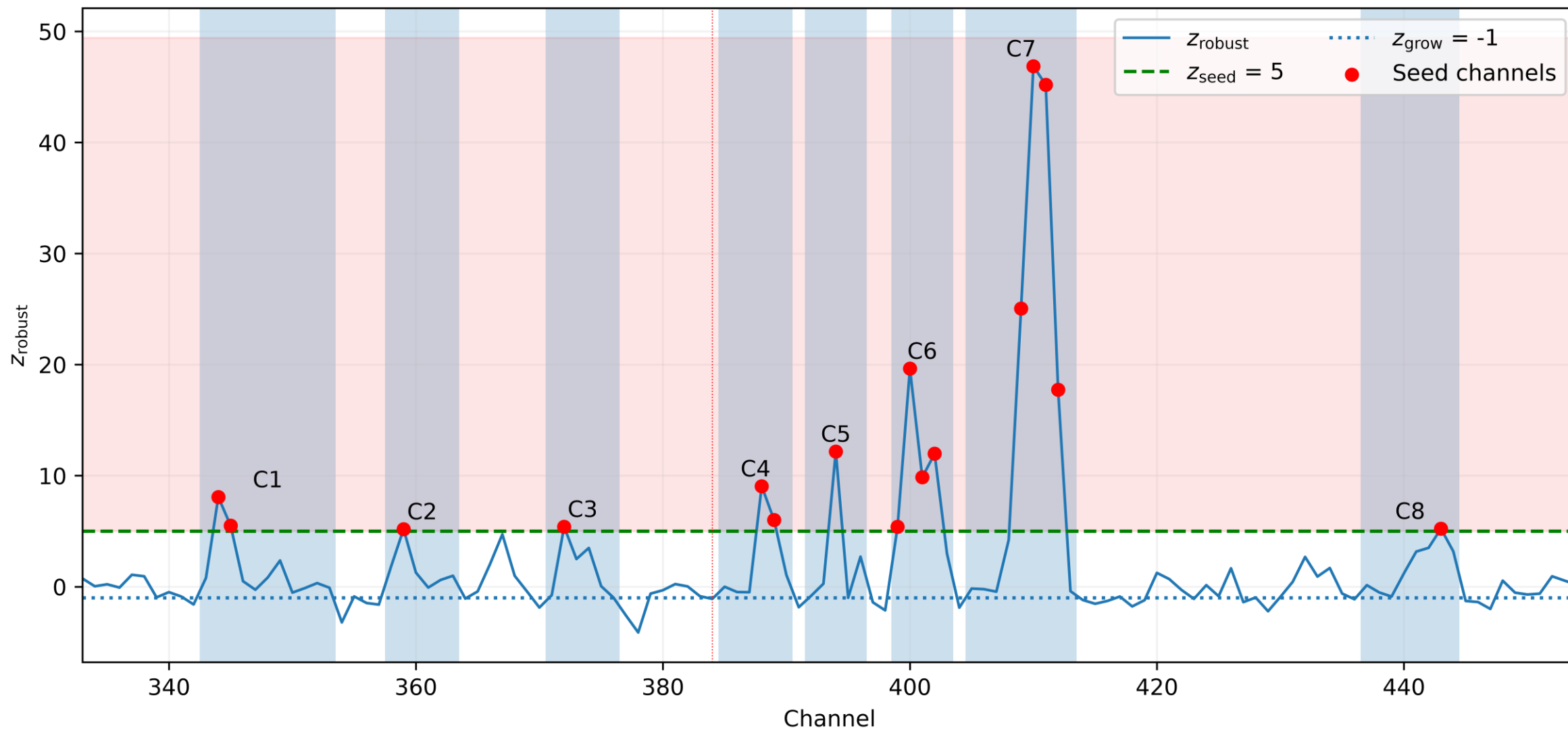
Event 18: Candidate Seeds → Candidate Clusters



1. Seed&Grow

II. Hit Reconstruction

Event 18: Candidate Seeds → Candidate Clusters

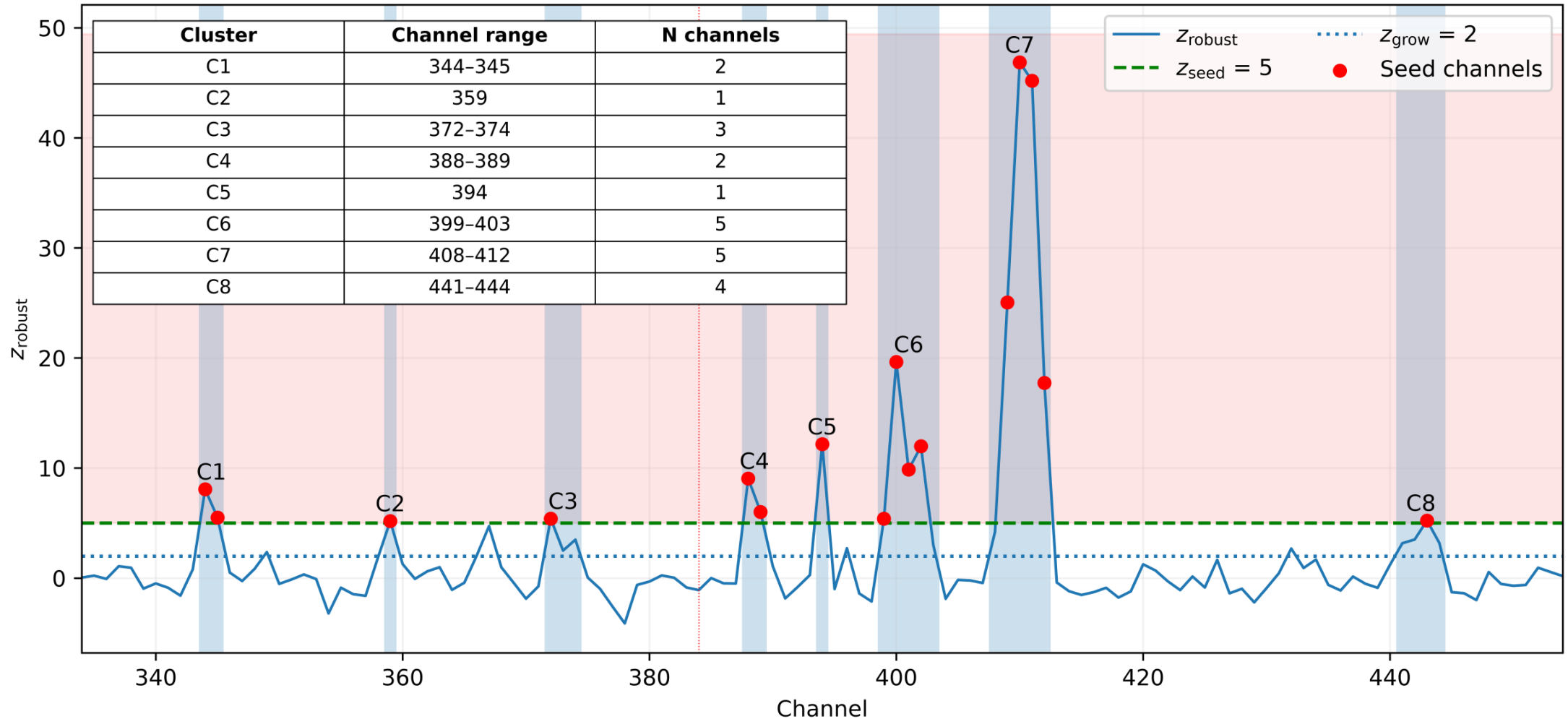


1. Seed&Grow

II. Hit Reconstruction

- The output is a set of contiguous candidate intervals $C_k = [L_k, R_k]$.

Event 18: Candidate Seeds → Candidate Clusters

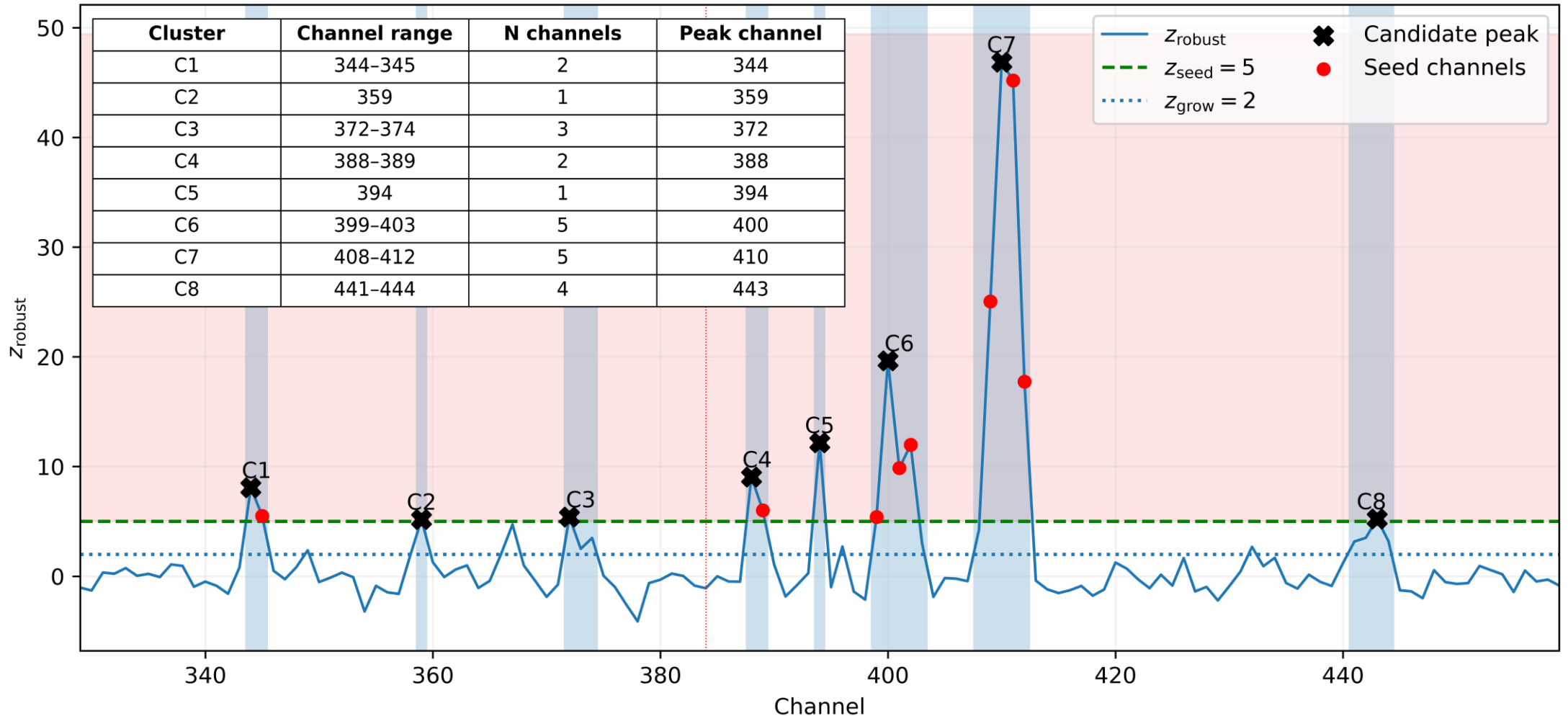


1. Seed&Grow

II. Hit Reconstruction

- Find the peak: $p_k = \arg \max_{j \in [L_k, R_k]} z_j^{\text{robust numerator}}$

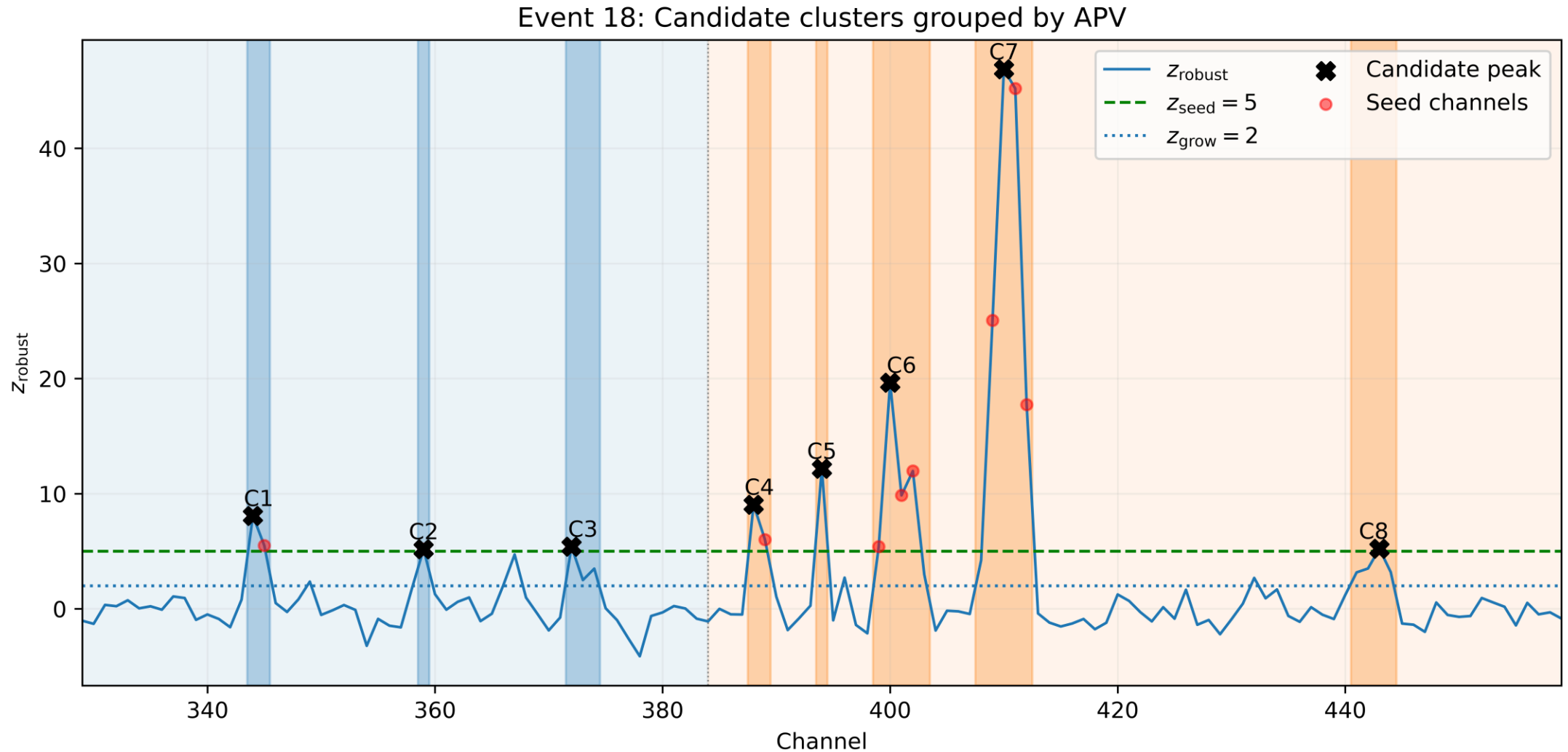
Event 18: Candidate Seeds → Candidate Clusters → Peaks



2. CrosstalkCleaning

II. Hit Reconstruction

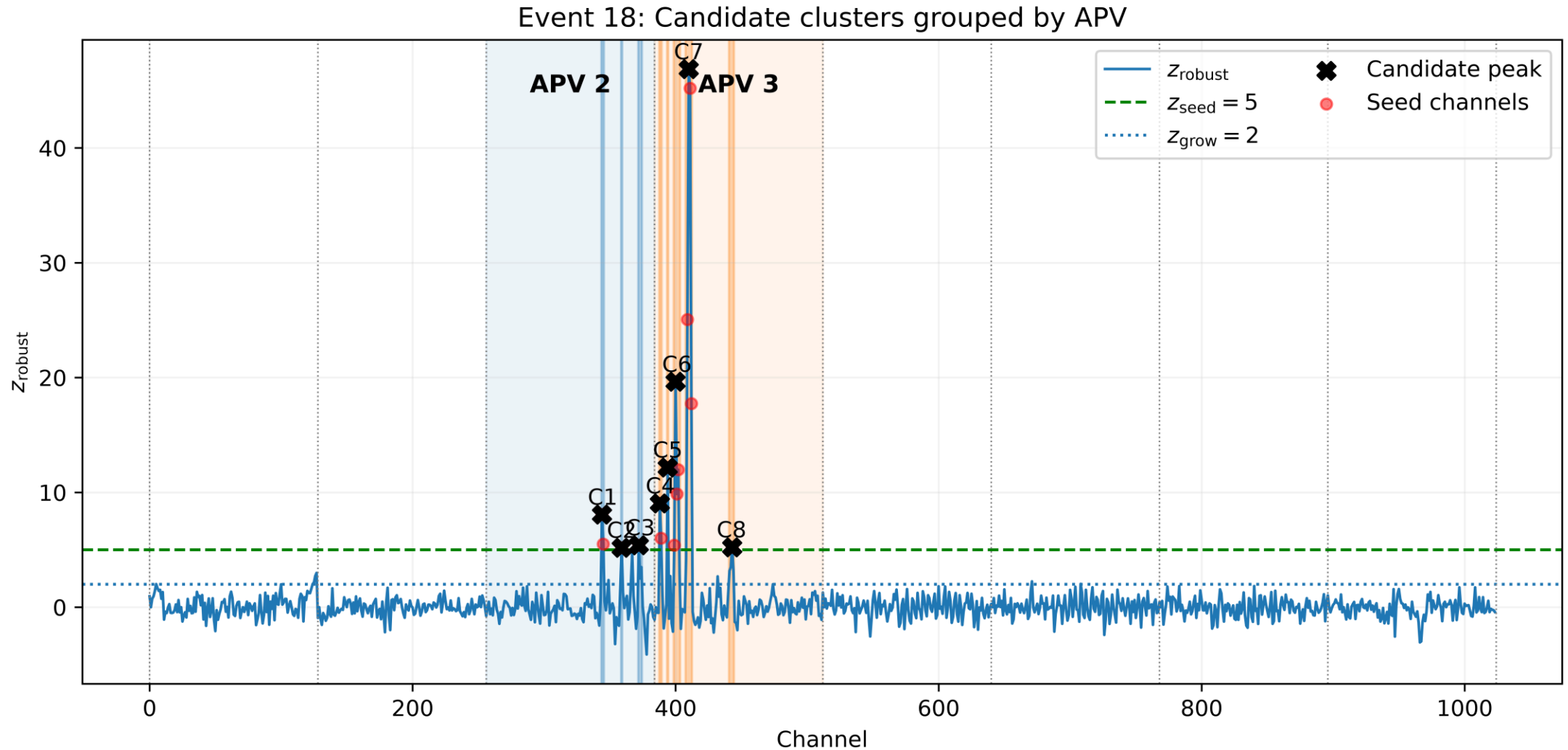
- Group clusters by APV



2. CrosstalkCleaning

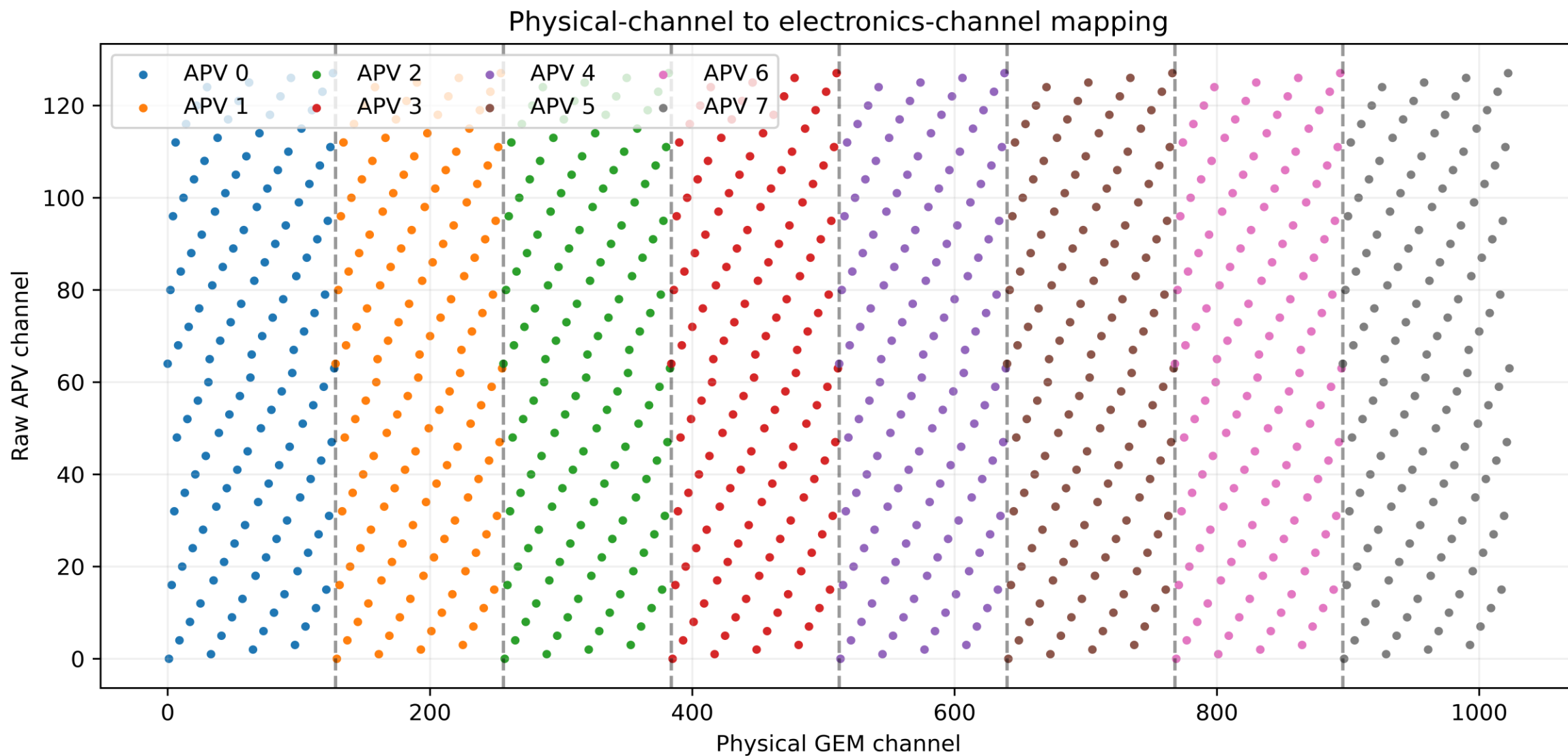
II. Hit Reconstruction

- Group clusters by APV



2. CrosstalkCleaning

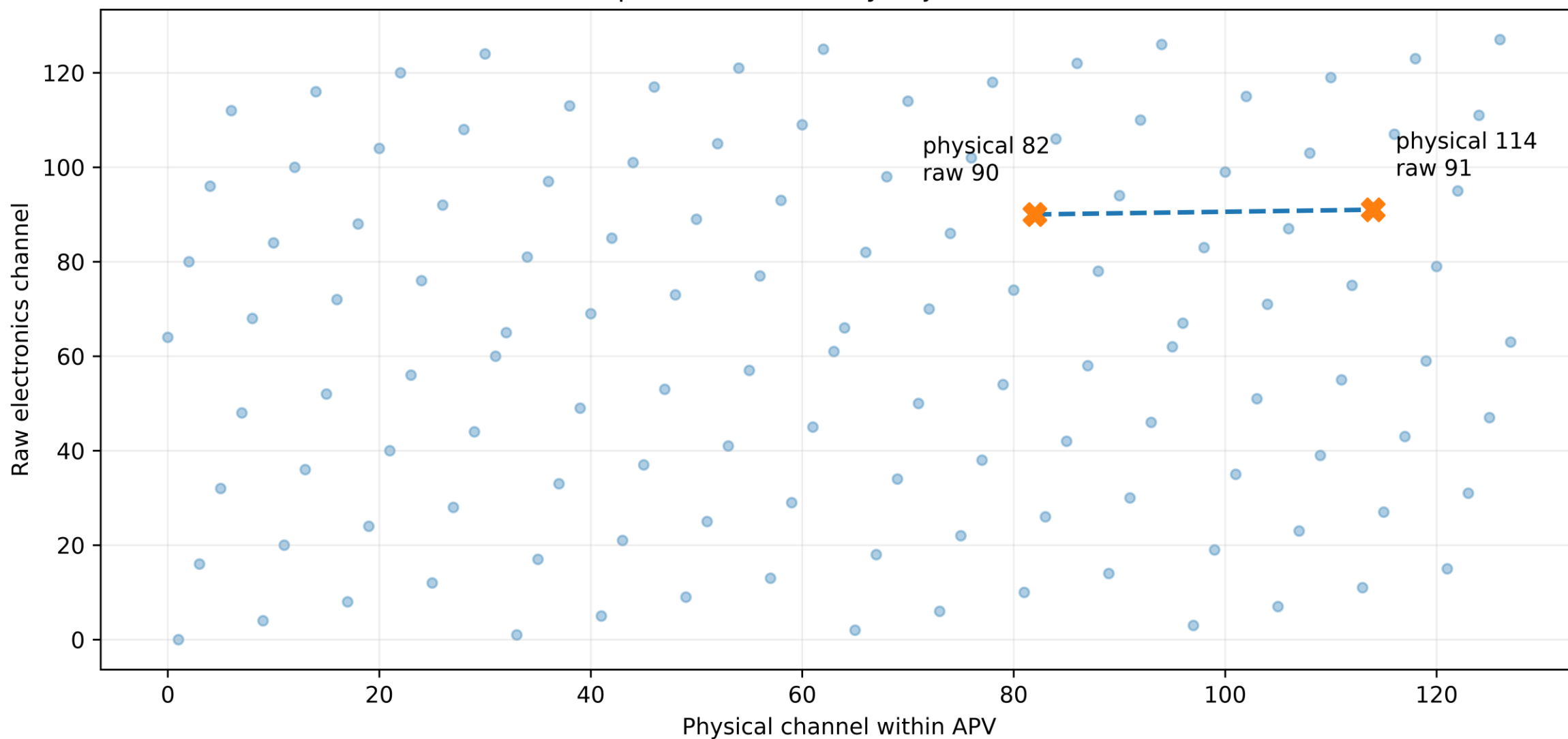
II. Hit Reconstruction



2. CrosstalkCleaning

II. Hit Reconstruction

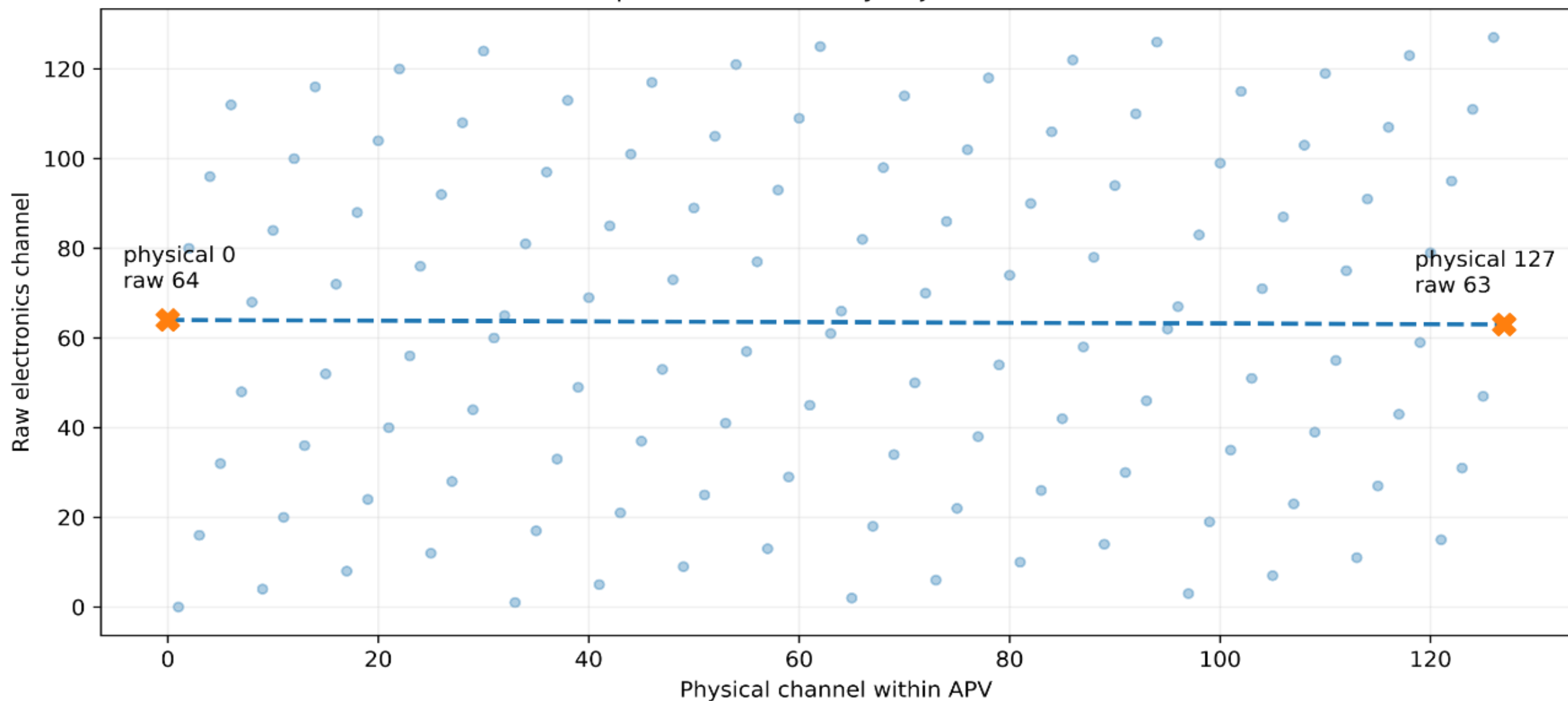
Example of electronically adjacent channels



2. CrosstalkCleaning

II. Hit Reconstruction

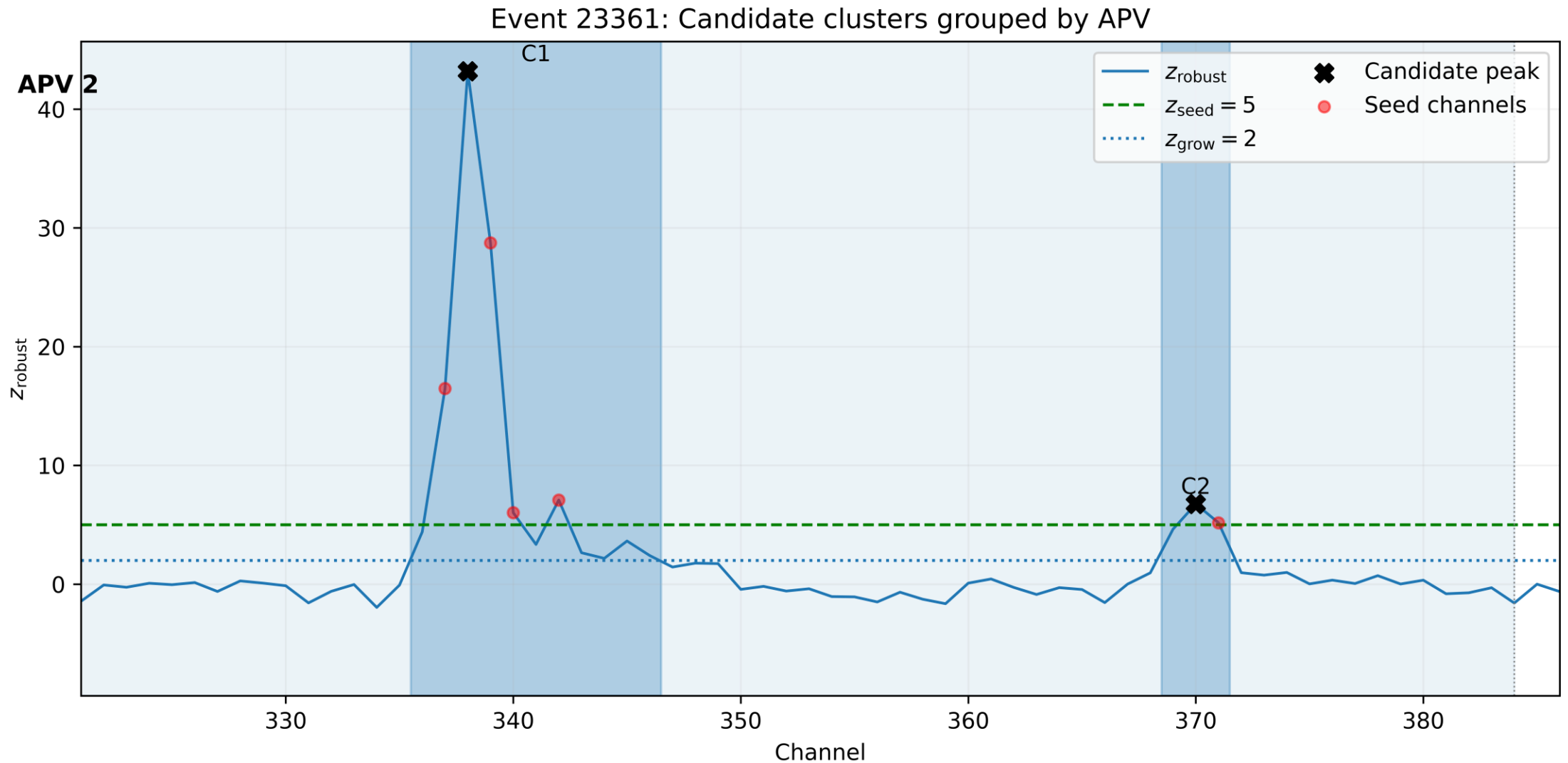
Example of electronically adjacent channels



2. CrosstalkCleaning

II. Hit Reconstruction

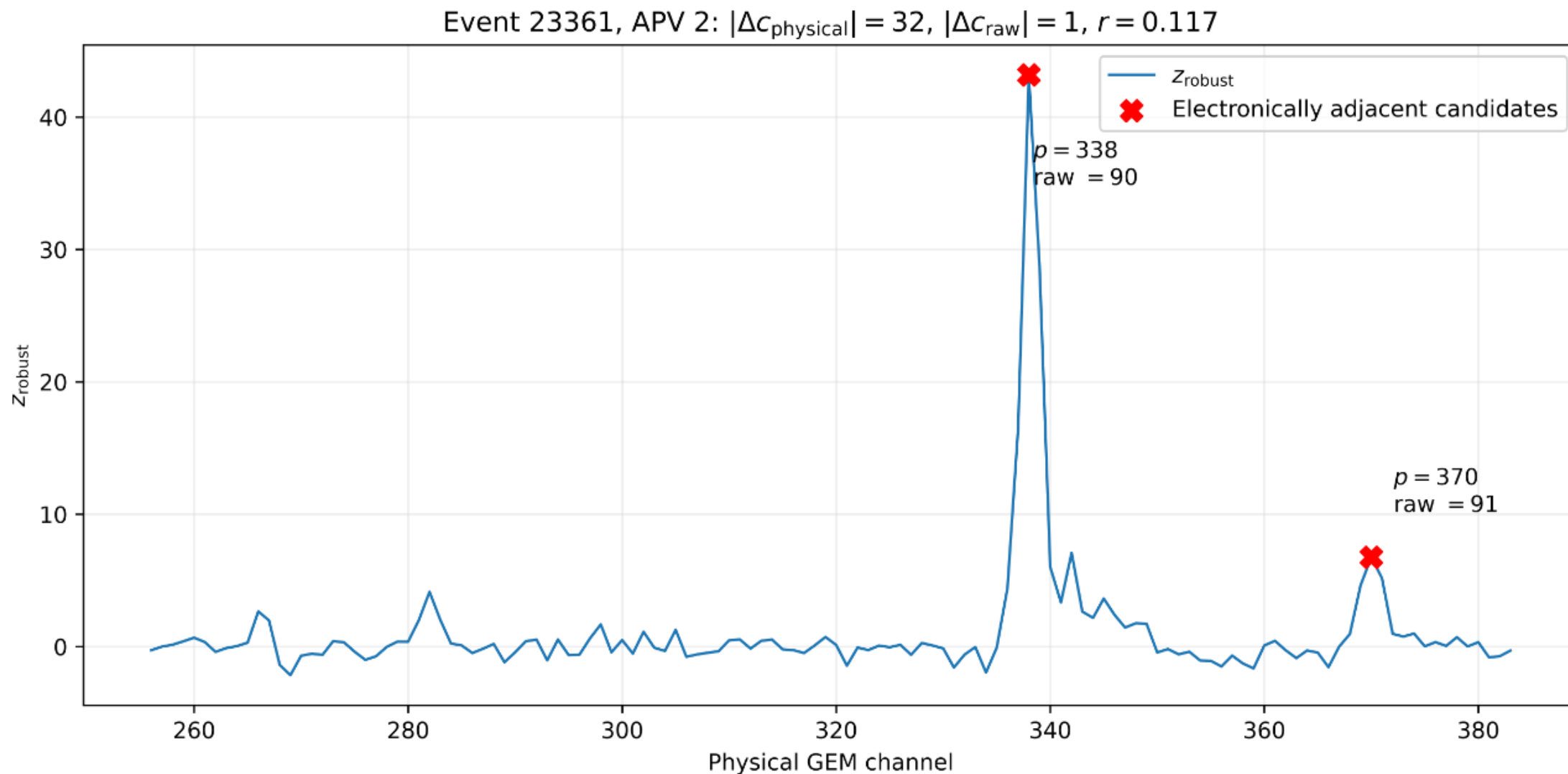
- Group clusters by APV



2. CrosstalkCleaning

II. Hit Reconstruction

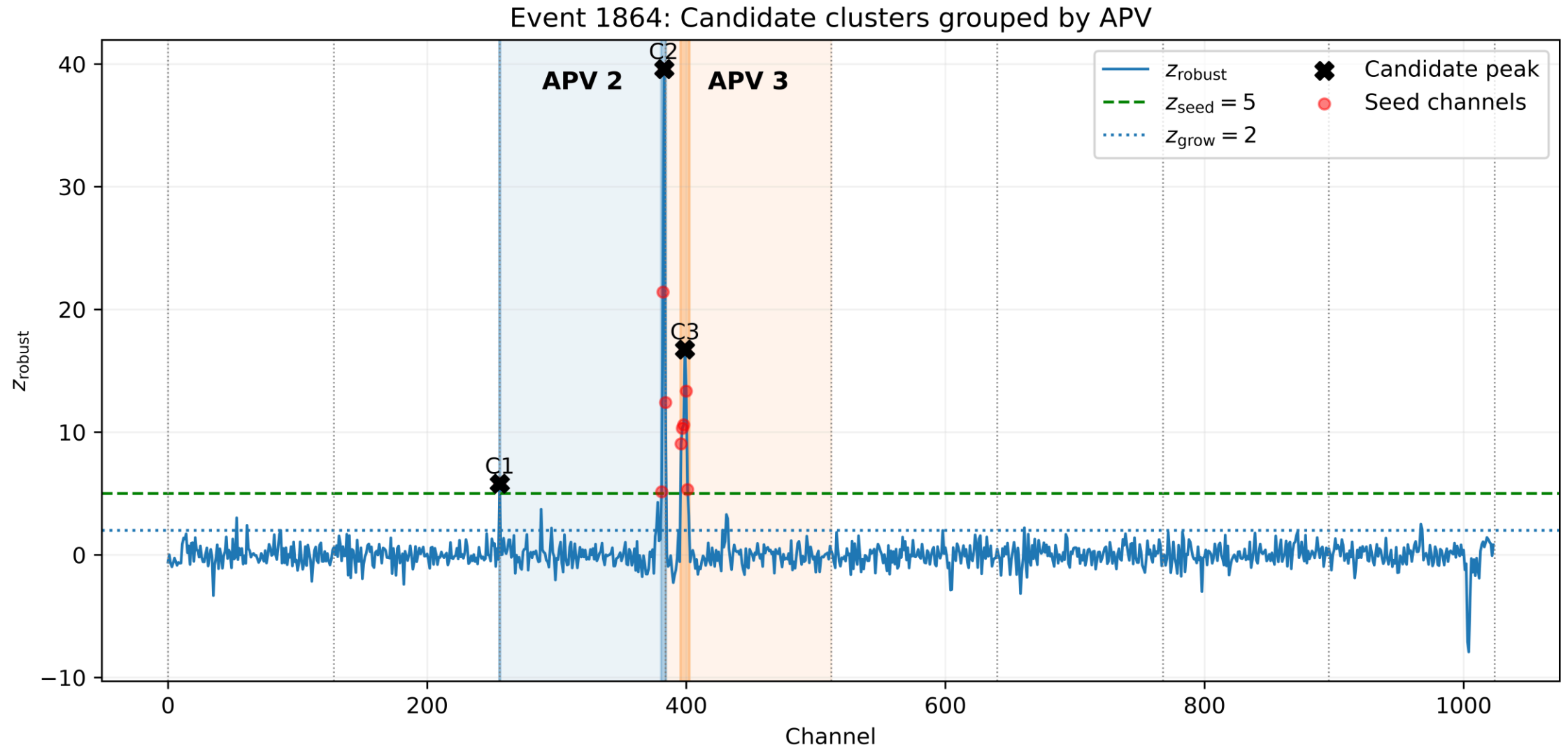
- Group clusters by APV



2. CrosstalkCleaning

II. Hit Reconstruction

- Group clusters by APV

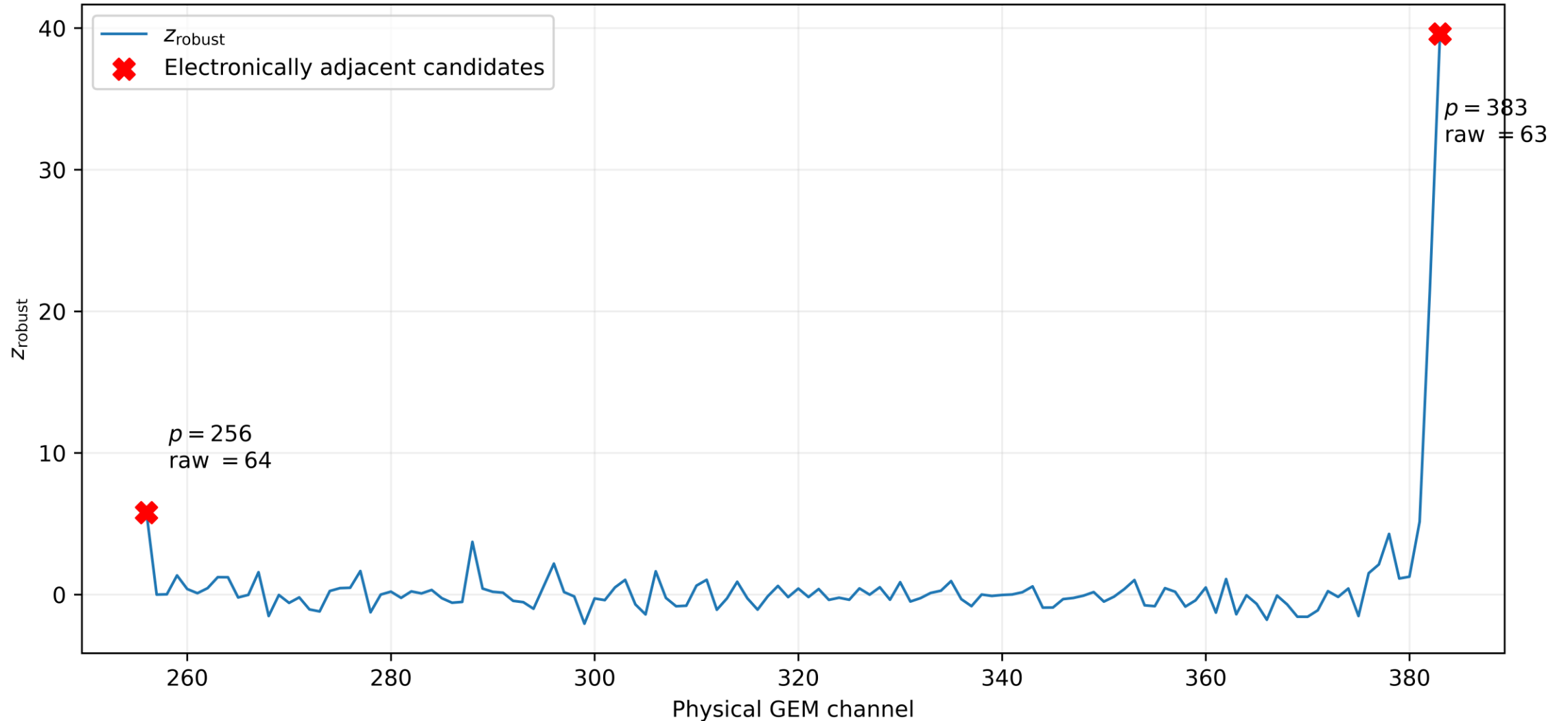


2. CrosstalkCleaning

II. Hit Reconstruction

- Group clusters by APV

Event 1864, APV 2: $|\Delta c_{\text{physical}}| = 127$, $|\Delta c_{\text{raw}}| = 1$, $r = 0.154$



2. CrosstalkCleaning Calibration II. Hit Reconstruction

- Group candidate clusters by APV.
- For every pair of candidates within the same APV, let A_1 and A_2 denote their peak amplitudes, where

$$A_k = s^{\text{hier}}(p_k).$$

- Define the amplitude ratio

$$\rho = \frac{\min(A_1, A_2)}{\max(A_1, A_2)}.$$

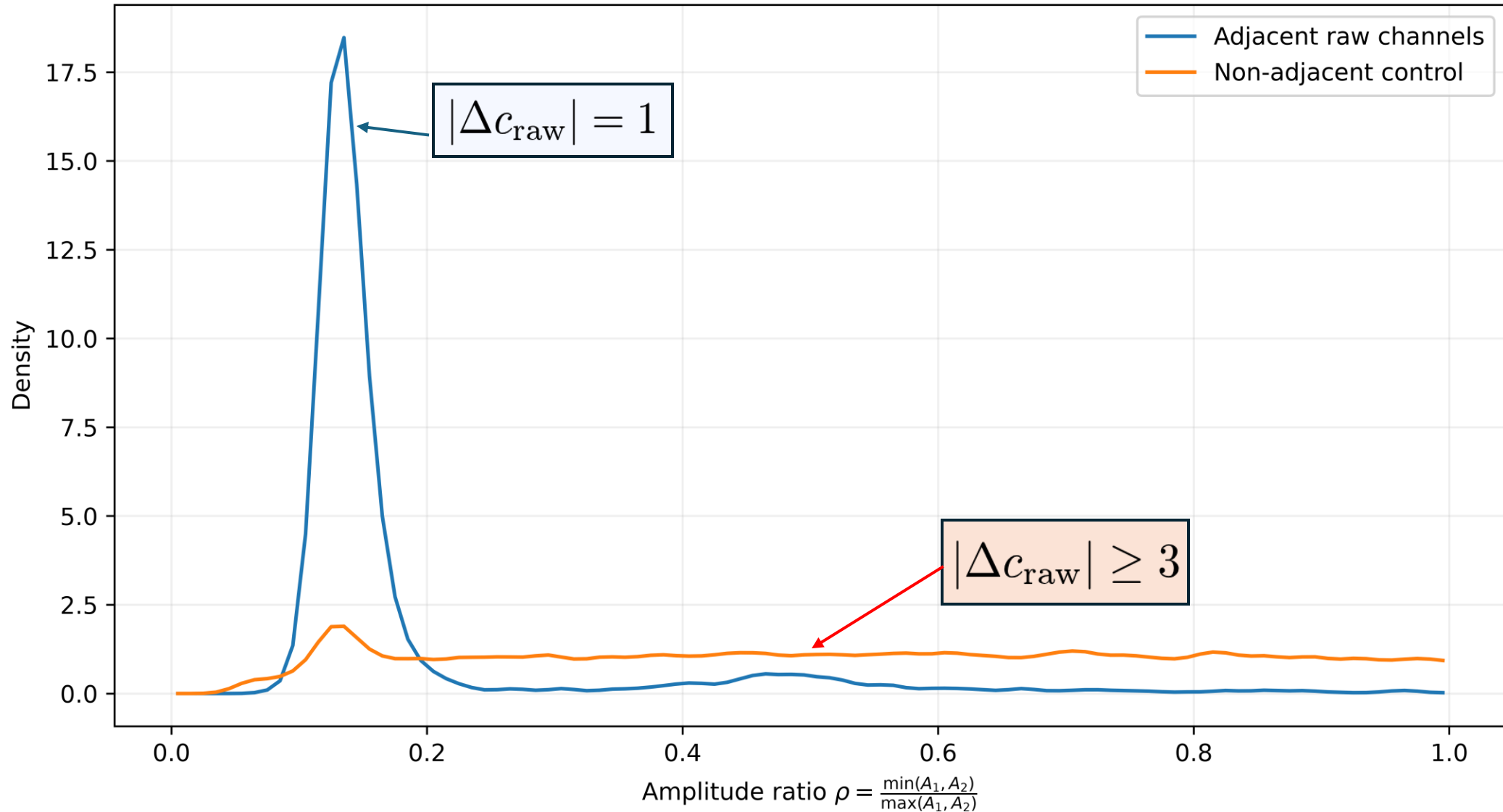
- Separate candidate pairs according to their raw-electronics channel separation:

$$|\Delta c_{\text{raw}}| = 1 \quad (\text{adjacent electronics channels})$$

and

$$|\Delta c_{\text{raw}}| \geq 3 \quad (\text{non-adjacent control}).$$

2. CrosstalkCleaning Calibration II. Hit Reconstruction



2. CrosstalkCleaning Calibration II. Hit Reconstruction

- Separate candidate pairs according to their raw-electronics channel separation:

$$|\Delta c_{\text{raw}}| = 1 \quad (\text{adjacent electronics channels})$$

and

$$|\Delta c_{\text{raw}}| \geq 3 \quad (\text{non-adjacent control}).$$

- The excess population is defined as

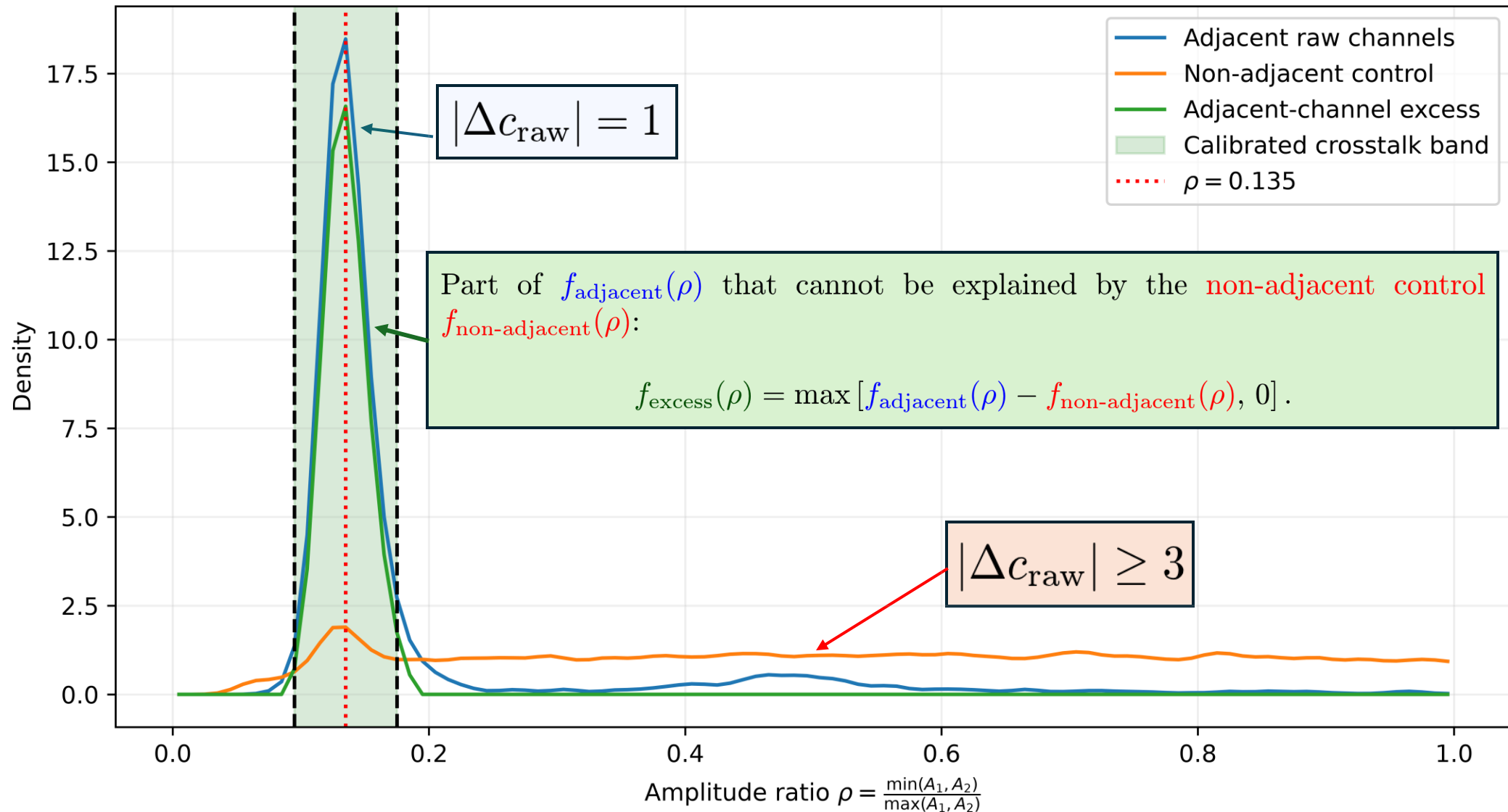
$$f_{\text{excess}}(\rho) = \max[f_{\text{adjacent}}(\rho) - f_{\text{control}}(\rho), 0].$$

- A narrow excess near

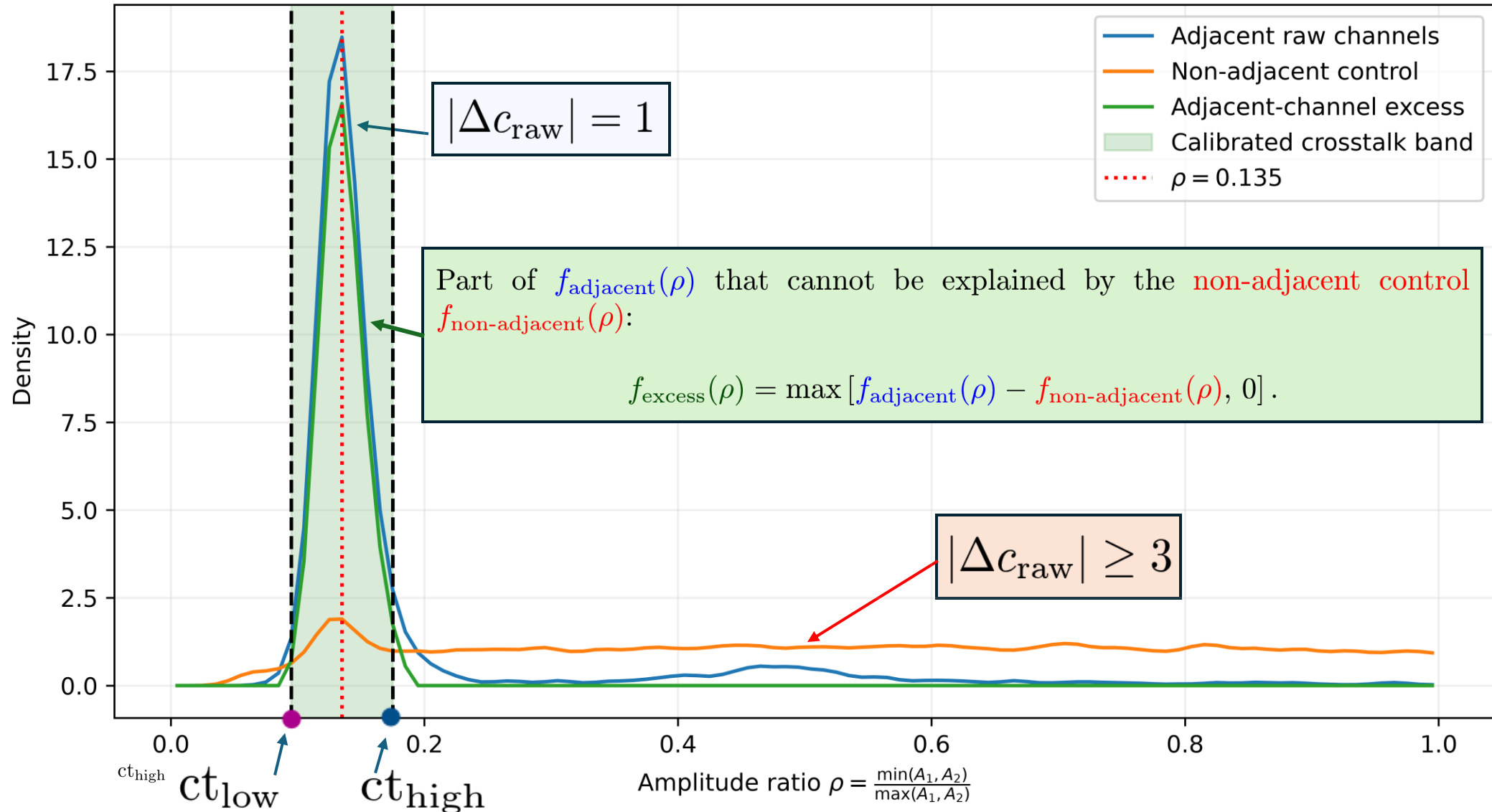
$$\rho \approx 0.135$$

indicates a characteristic amplitude ratio associated with electronics crosstalk.

2. CrosstalkCleaning Calibration II. Hit Reconstruction



2. CrosstalkCleaning Calibration II. Hit Reconstruction



Crosstalk Removal Rule

For each pair of candidate clusters within the same APV:

$$|\Delta c_{\text{raw}}| = 1$$

and

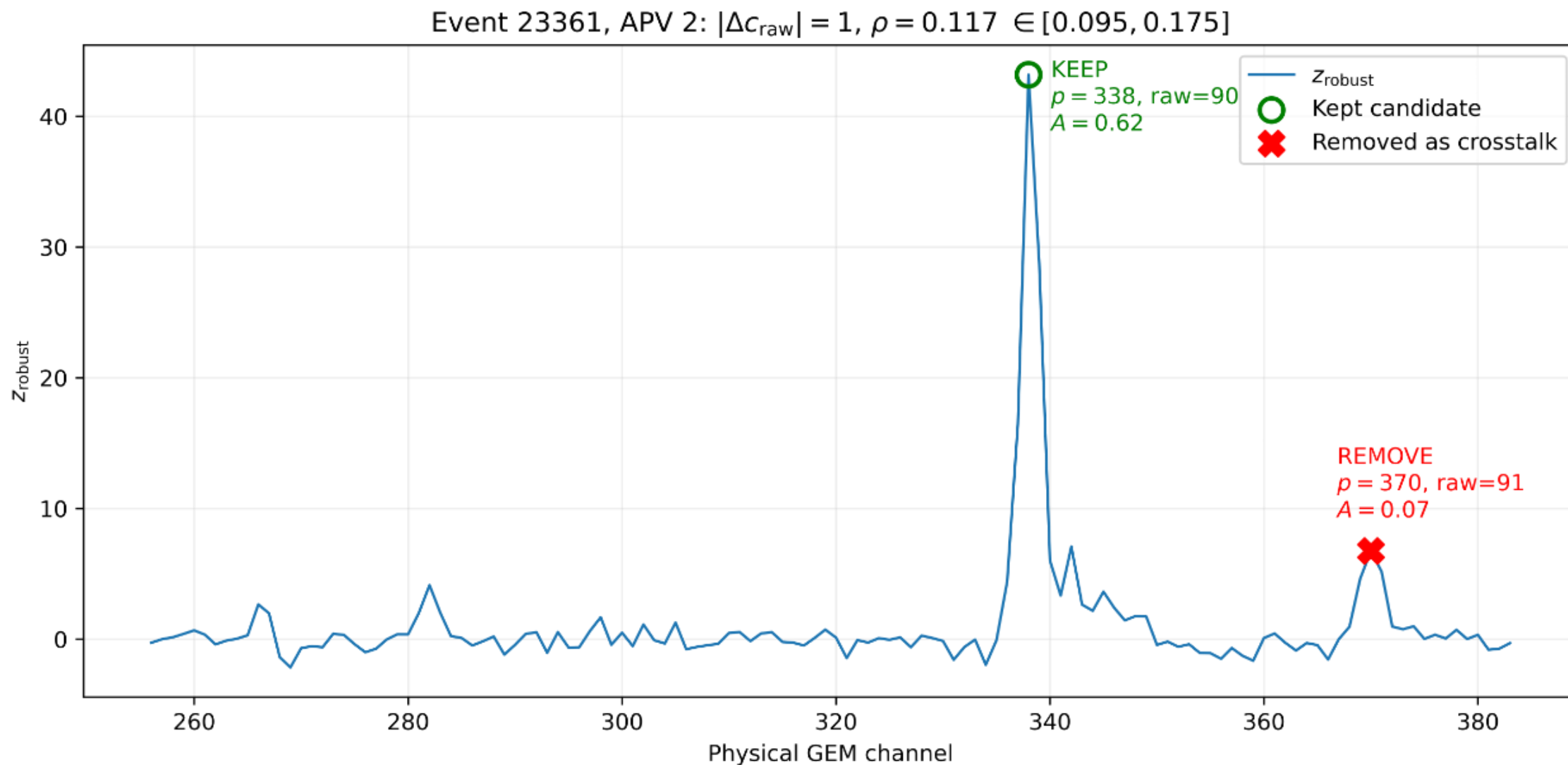
$$ct_{\text{low}} \leq \rho = \frac{\min(A_1, A_2)}{\max(A_1, A_2)} \leq ct_{\text{high}}.$$

If both conditions are satisfied, the weaker candidate is classified as an electronics-crosstalk echo:

$$A_{\text{weak}} \rightarrow \text{remove}, \quad A_{\text{strong}} \rightarrow \text{keep}.$$

2. CrosstalkCleaning

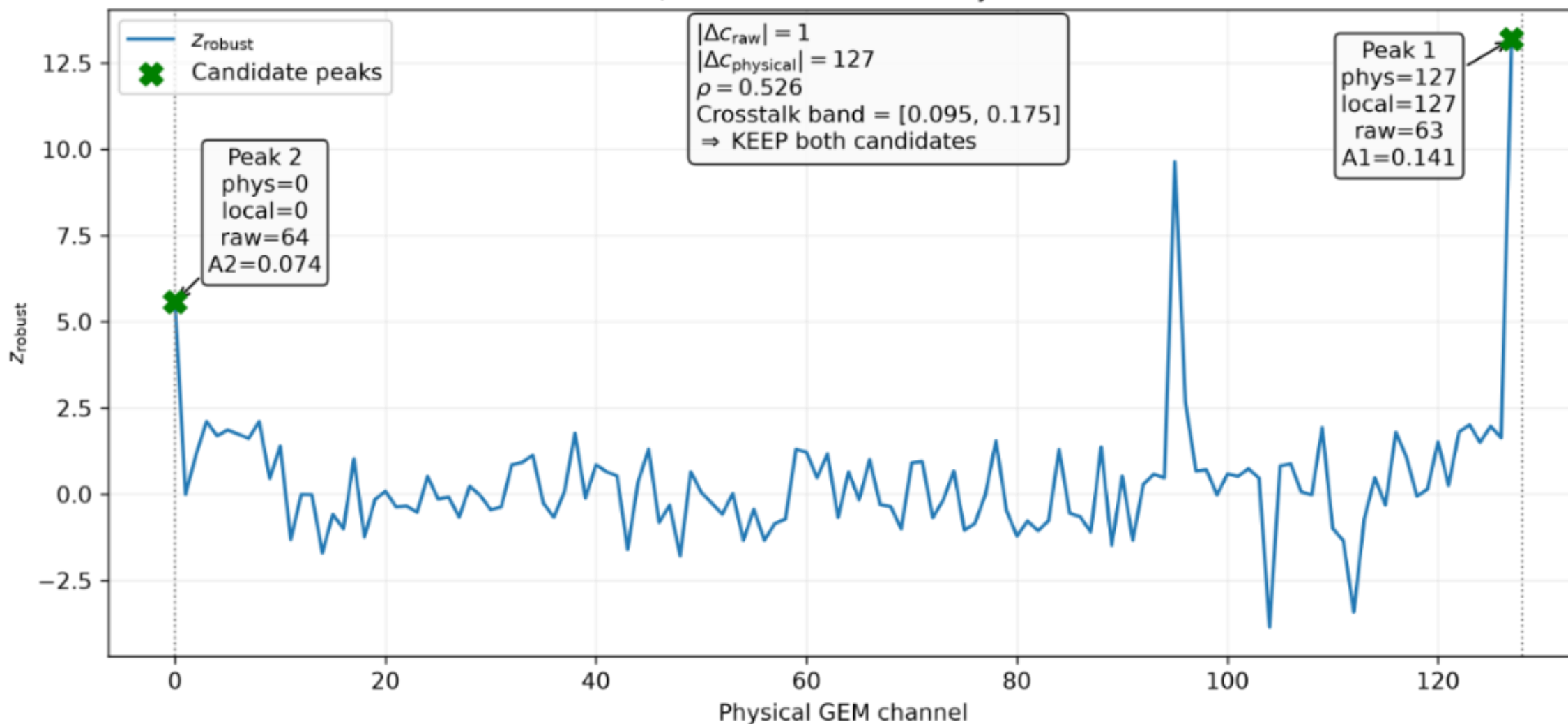
II. Hit Reconstruction



2. CrosstalkCleaning

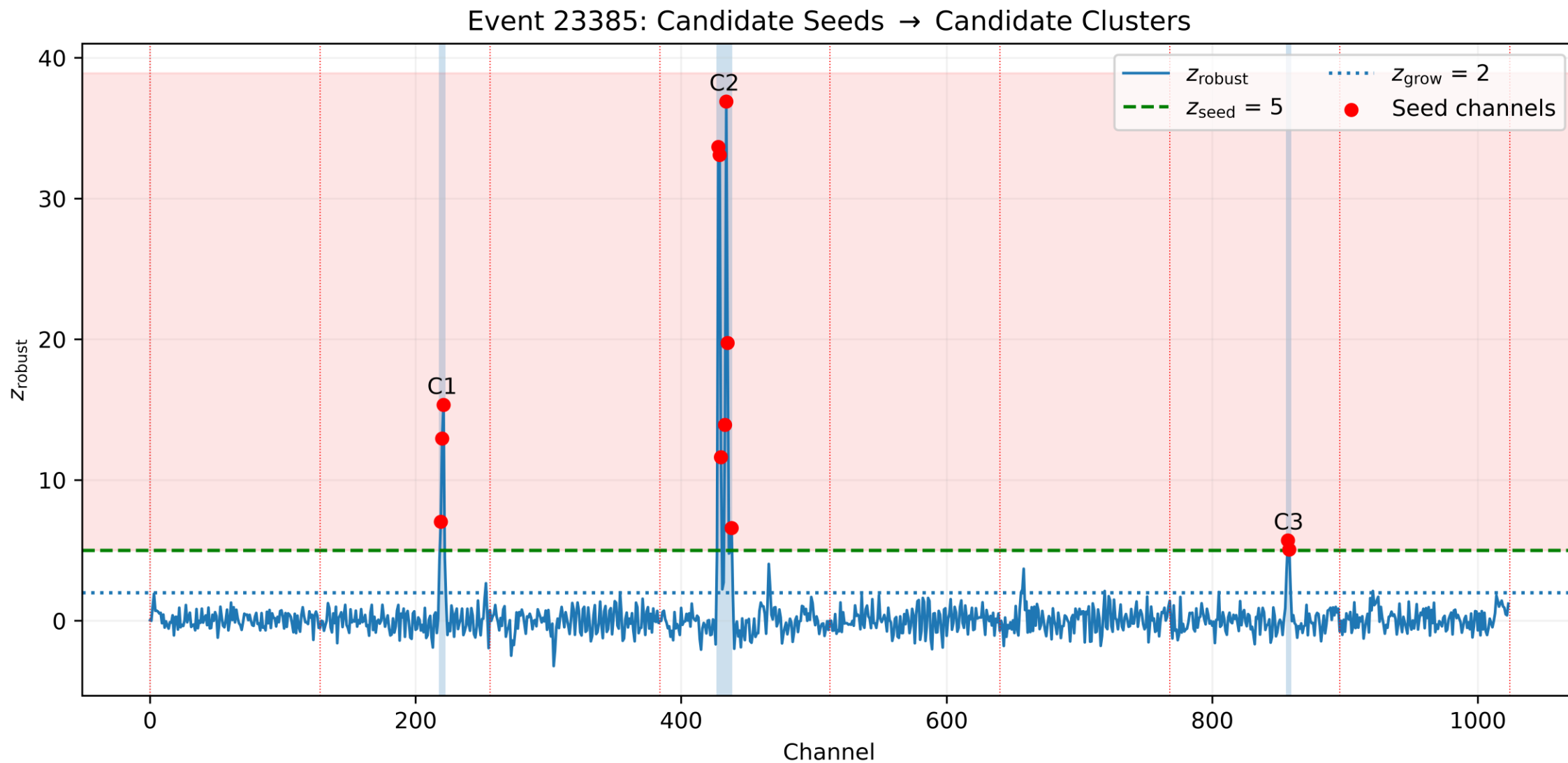
II. Hit Reconstruction

Event 4837, APV 0: Not removed by crosstalk rule



4. PeakSplitting

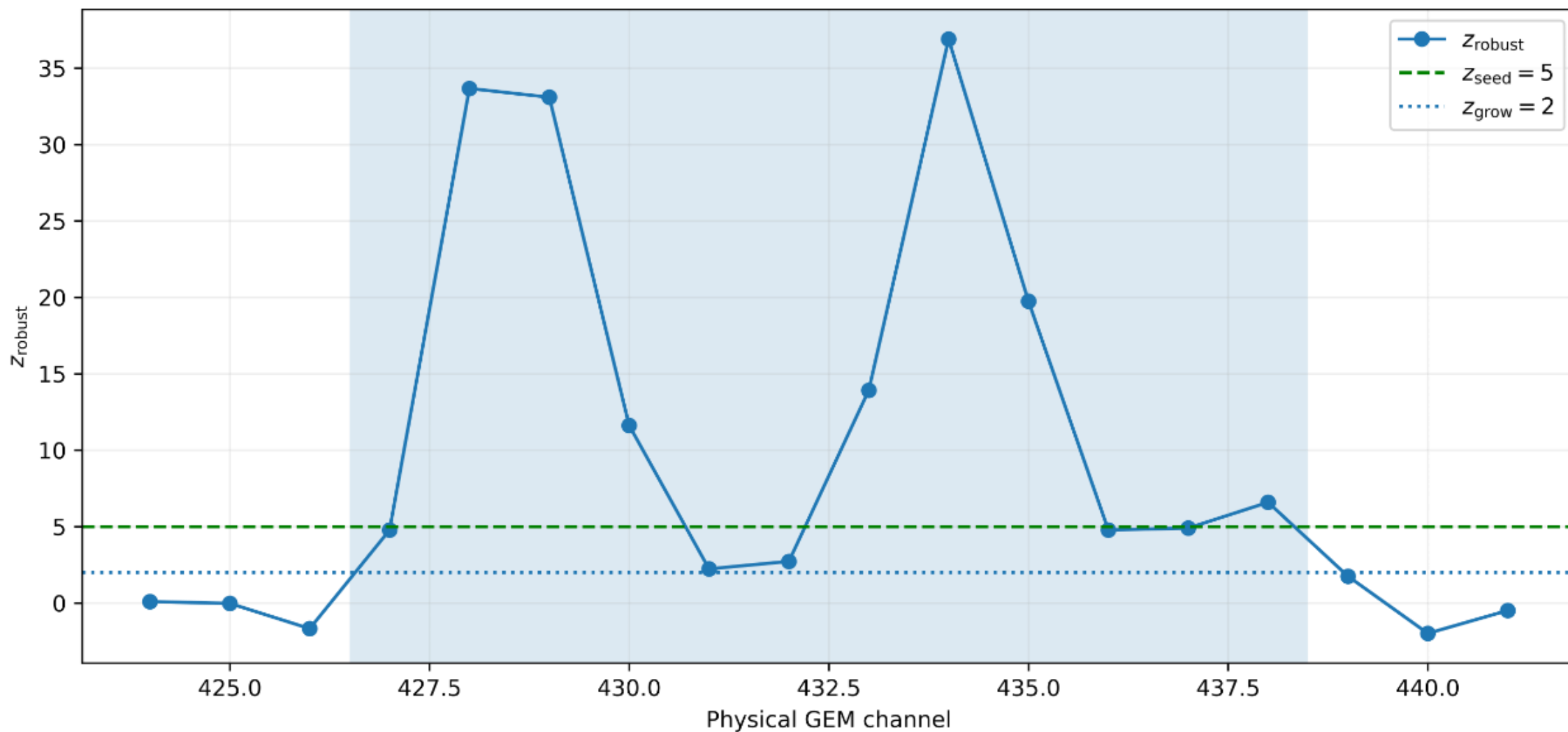
II. Hit Reconstruction



4. PeakSplitting

II. Hit Reconstruction

Event 23385: Candidate Cluster #2



4. PeakSplitting

After crosstalk removal, a surviving seed-and-grow candidate may still contain more than one physical hit.

- For each surviving candidate

$$C_k = [L_k, R_k],$$

we inspect the amplitude-like signal

$$s_j^{\text{hier}} = \text{signal}_{\text{hier}}[j].$$

- We search for two local maxima

$$p_1, p_2 \in C_k$$

and define

$$A_i = s_{p_i}^{\text{hier}}.$$

II. Hit Reconstruction

- The candidate is considered splittable if

$$|p_2 - p_1| \geq d_{\min},$$

$$\frac{\min(A_1, A_2)}{\max(A_1, A_2)} \geq r_{\min},$$

and

$$\frac{V}{\min(A_1, A_2)} \leq v_{\max},$$

where V is the minimum signal between the two peaks.

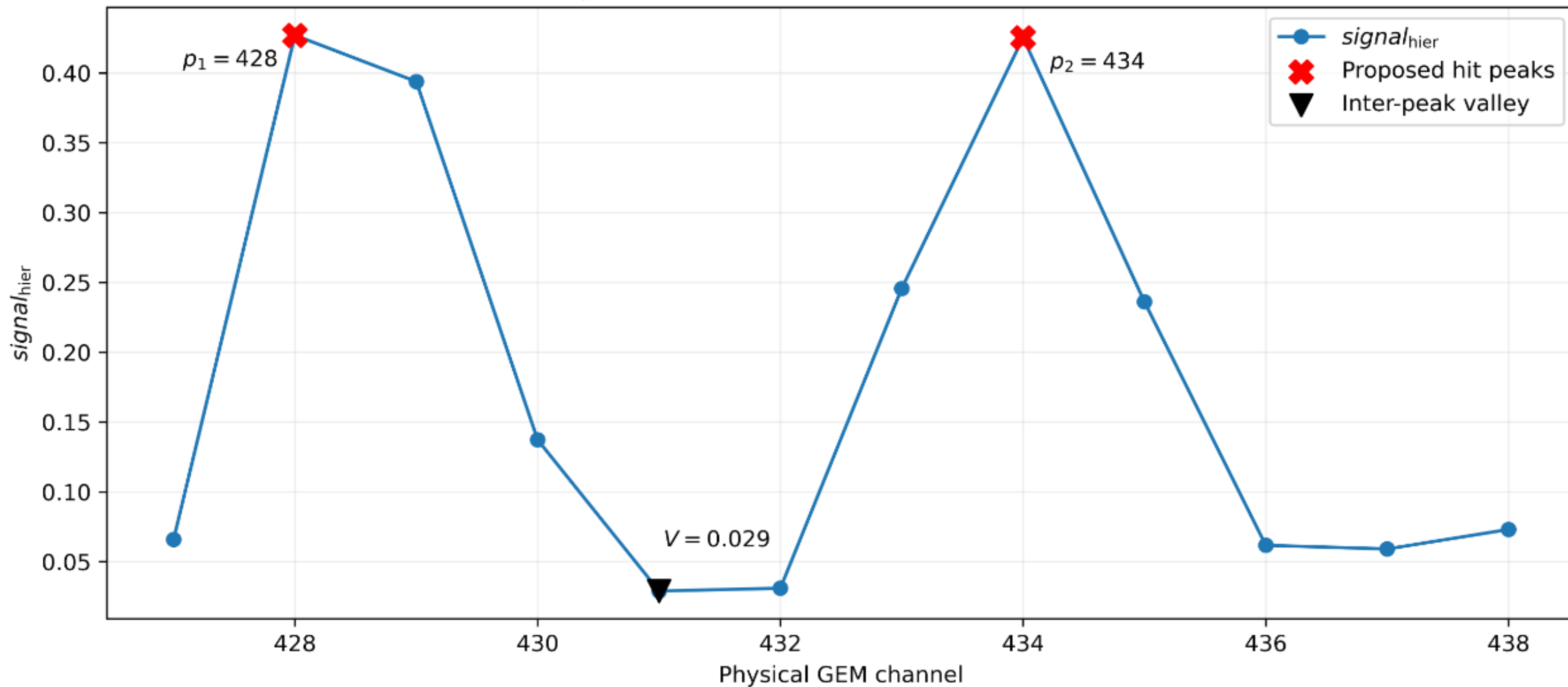
- For the current algorithm:

$$d_{\min} = 5, \quad r_{\min} = 0.30, \quad v_{\max} = 0.30.$$

4. PeakSplitting

II. Hit Reconstruction

Event 14942: Candidate [427,438] → two-hit split
 $\Delta p = 6$, $A_{\min}/A_{\max} = 0.996$, $V/A_{\min} = 0.068$



5. HitPosition

II. Hit Reconstruction

- Each surviving hit is associated with a dominant physical channel p_k .
- For a 3-strip window centered on p_k ,

$$j \in \{p_k - 1, p_k, p_k + 1\},$$

we use the positive part of the amplitude-like signal,

$$w_j = \max(s_j^{\text{hier}}, 0).$$

- The reconstructed hit position is then

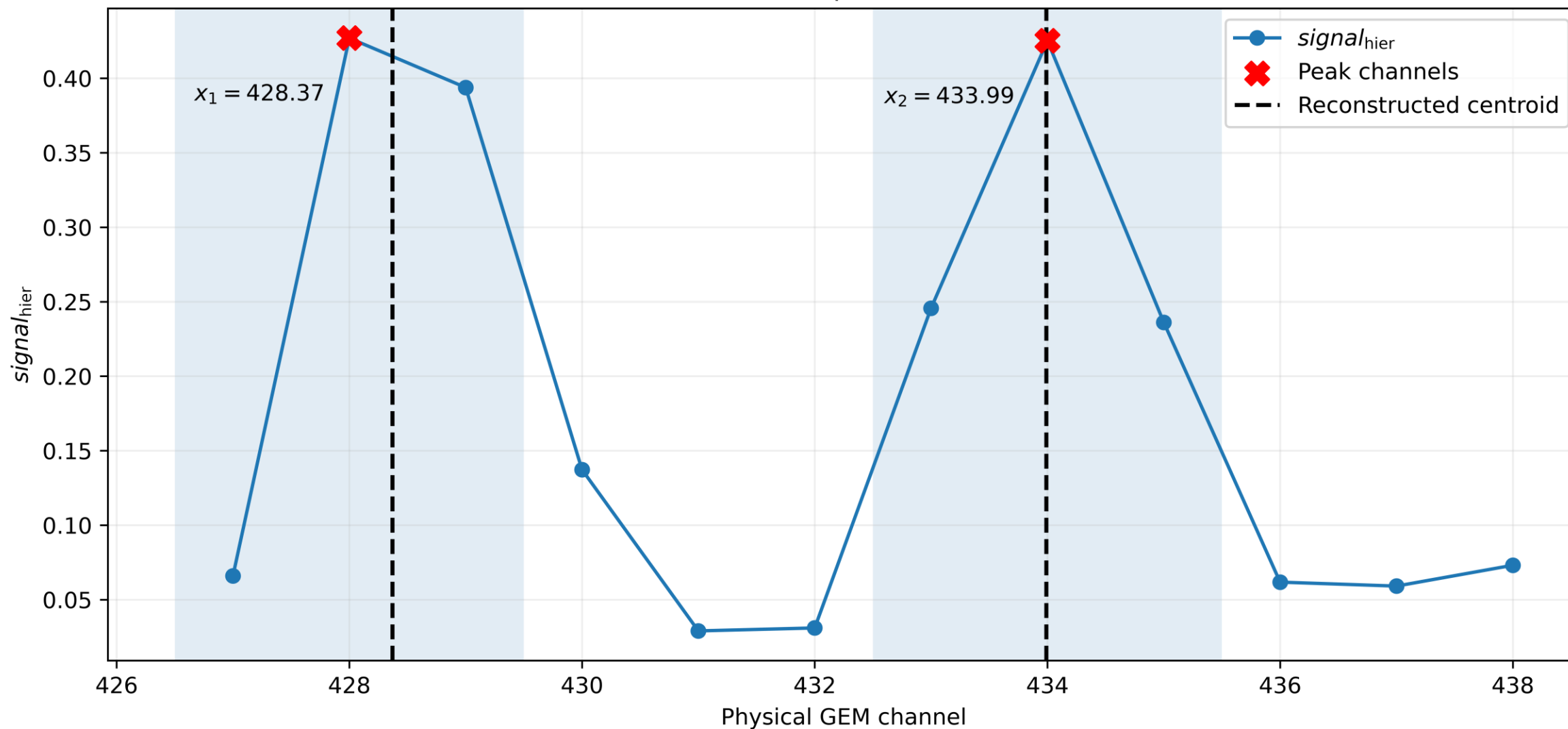
$$x_k = \frac{\sum_j j w_j}{\sum_j w_j}$$

provided that the total weight is nonzero.

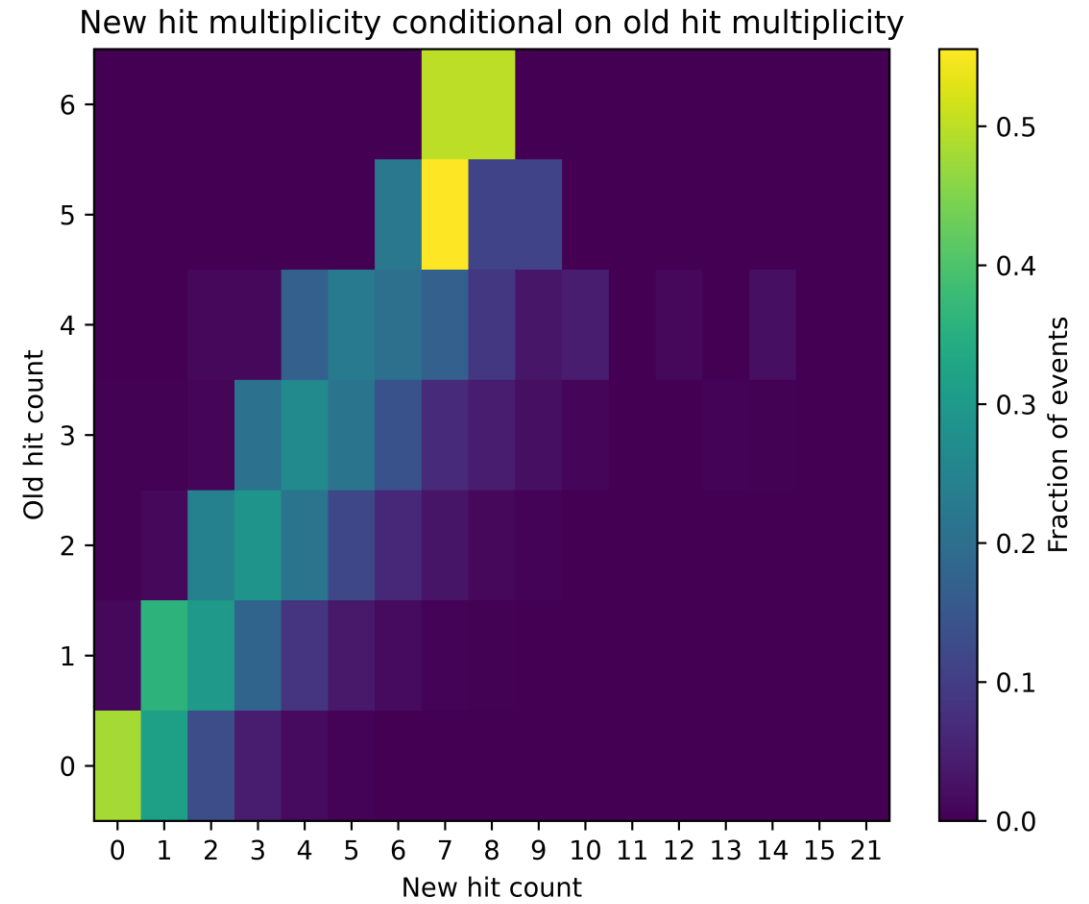
5. HitPosition

II. Hit Reconstruction

Final sub-channel hit-position reconstruction

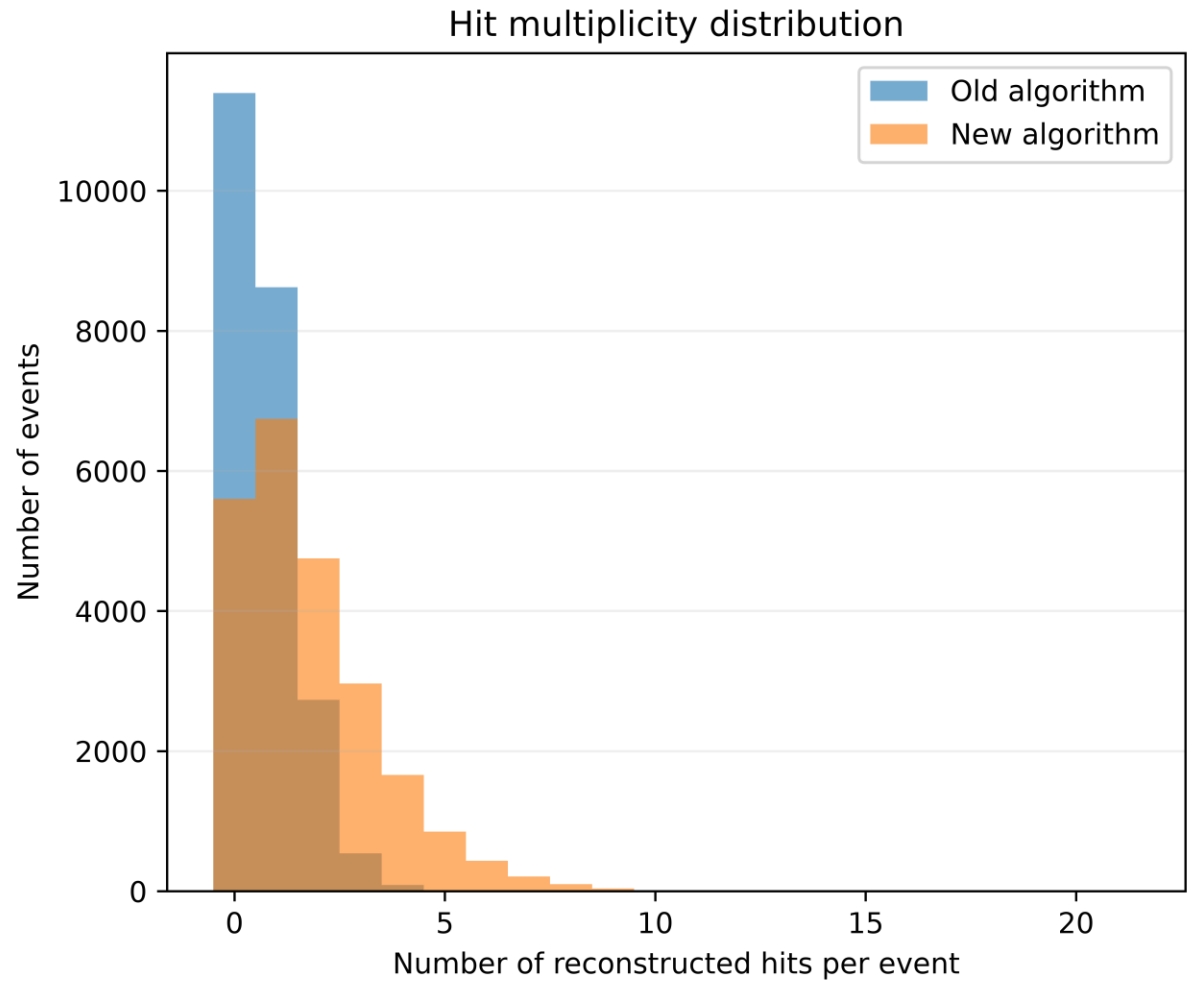
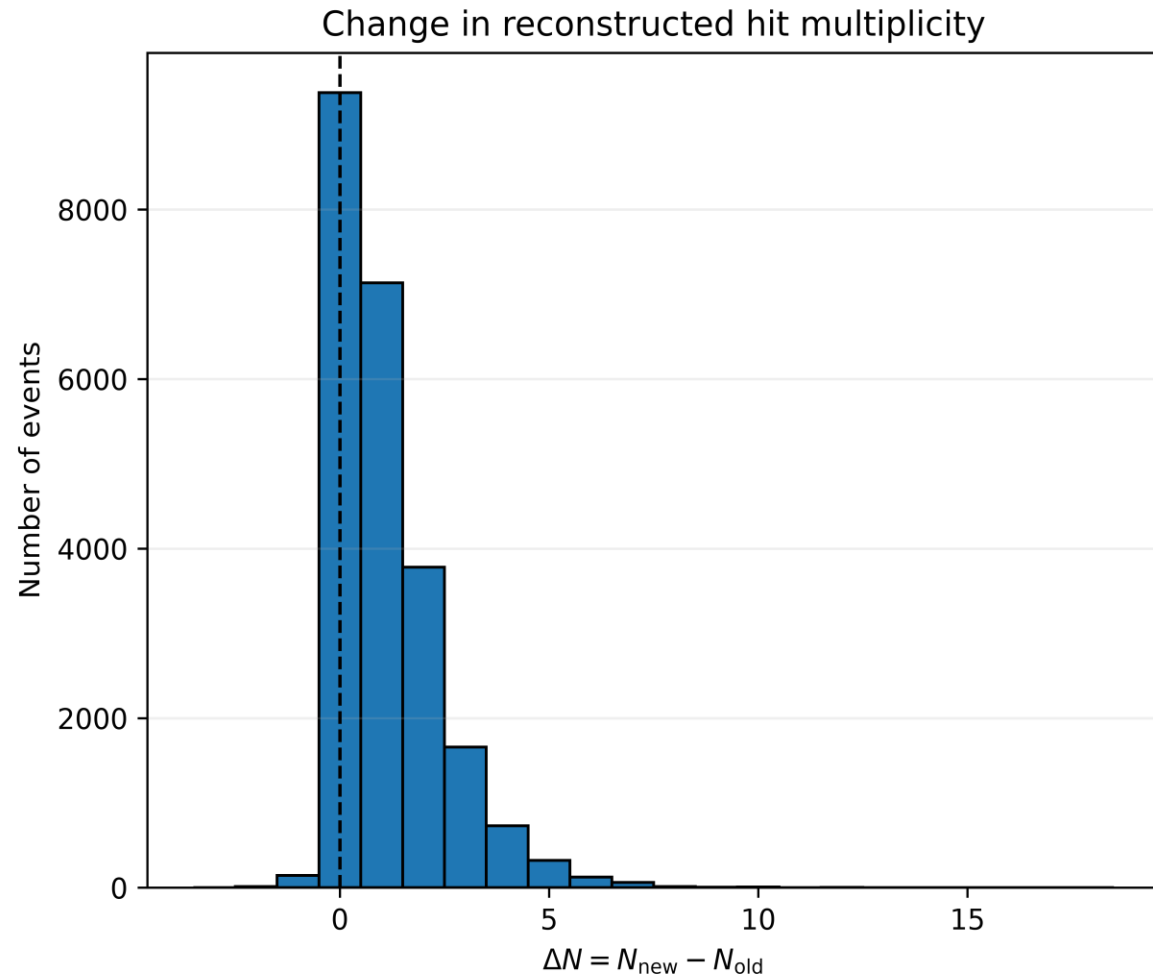


GEM0-X



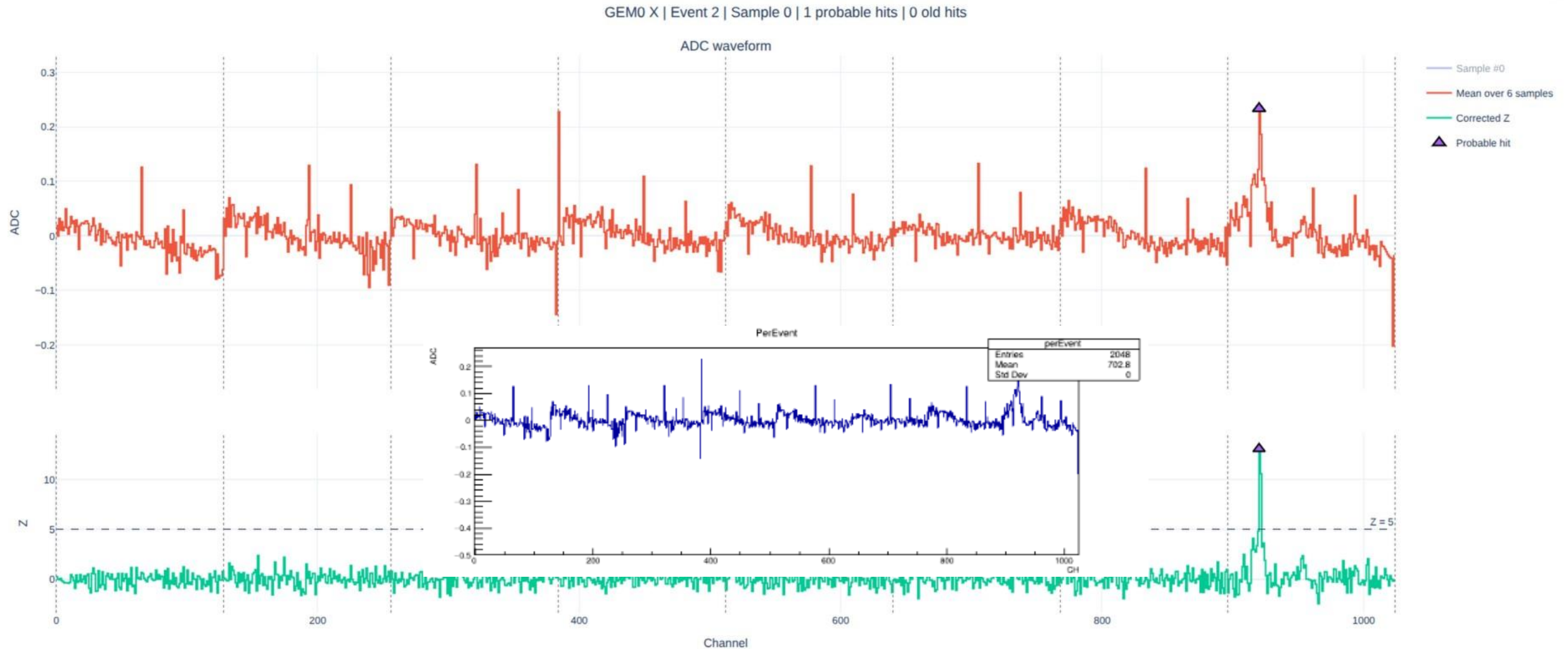
Results: Comparison

GEM0-X



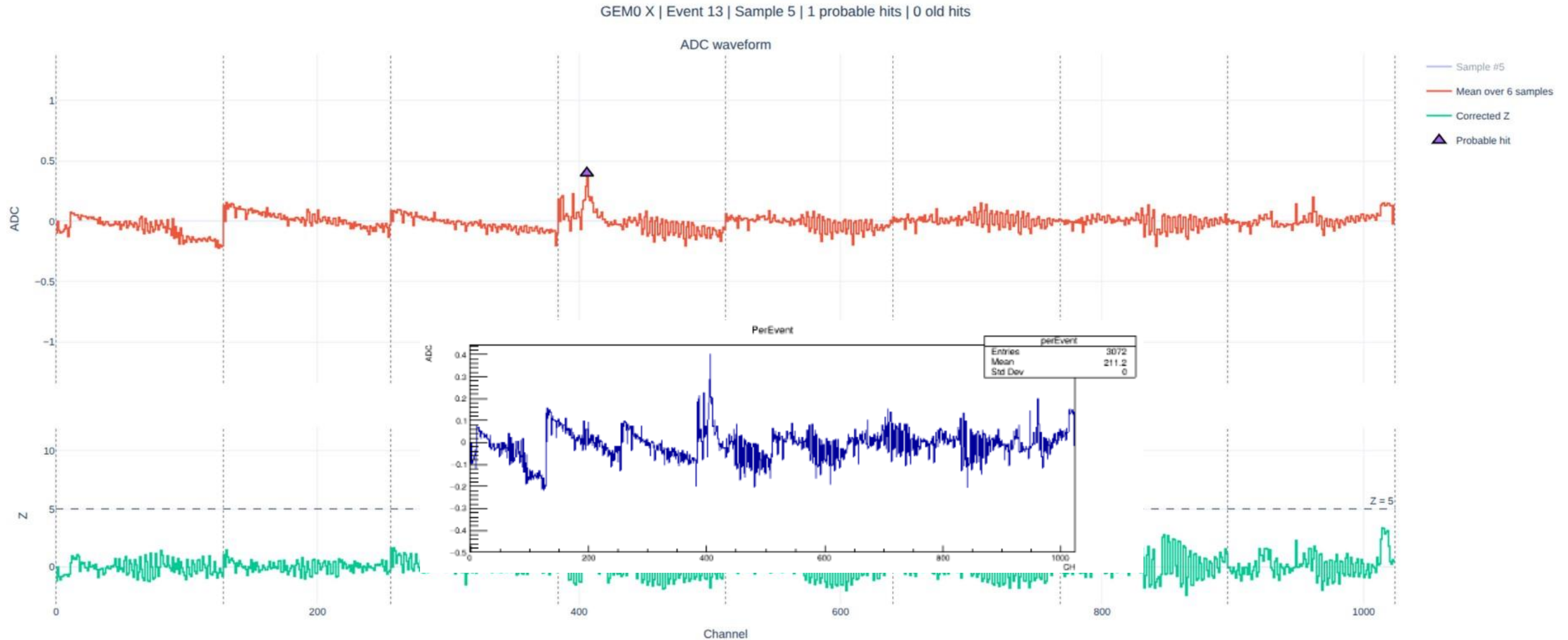
Results: New ✓ Old ✗

GEM0-X



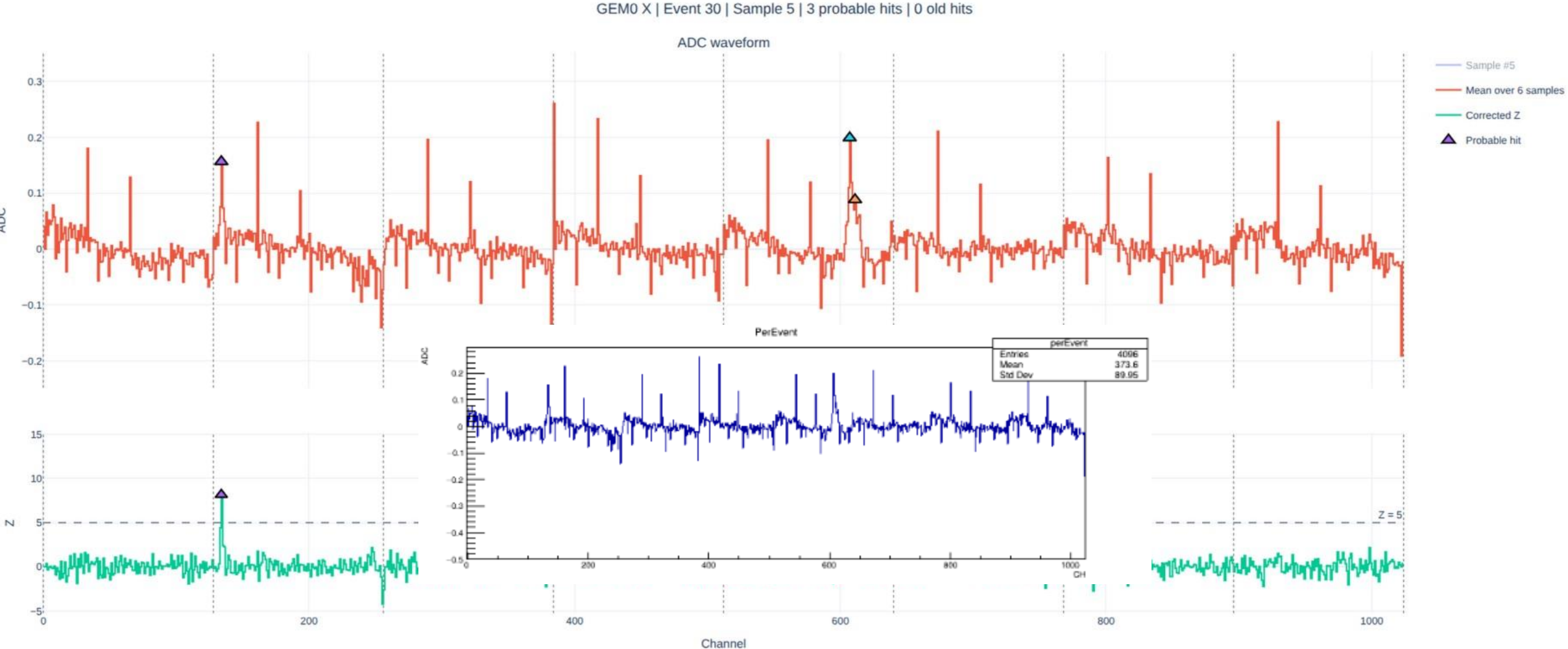
Results: New ✓ Old ✗

GEM0-X



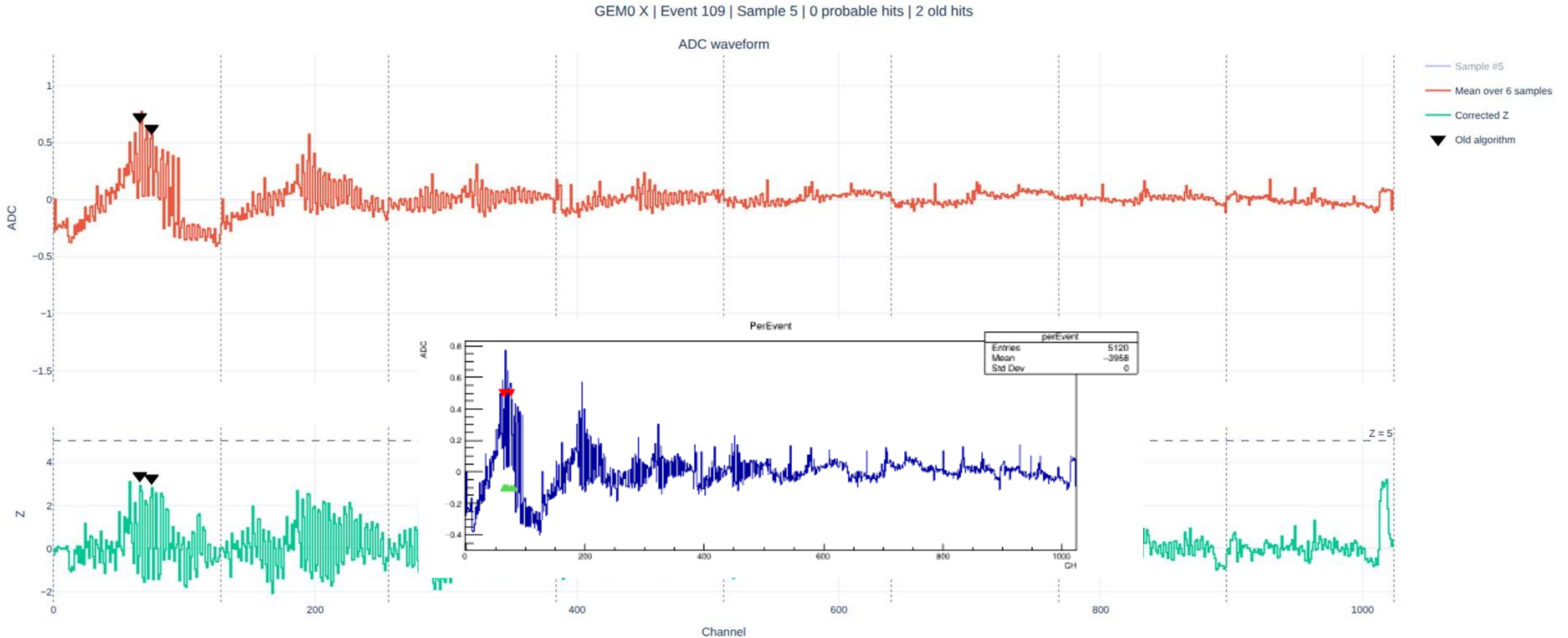
Results: New ✓ Old ✗

GEM0-X



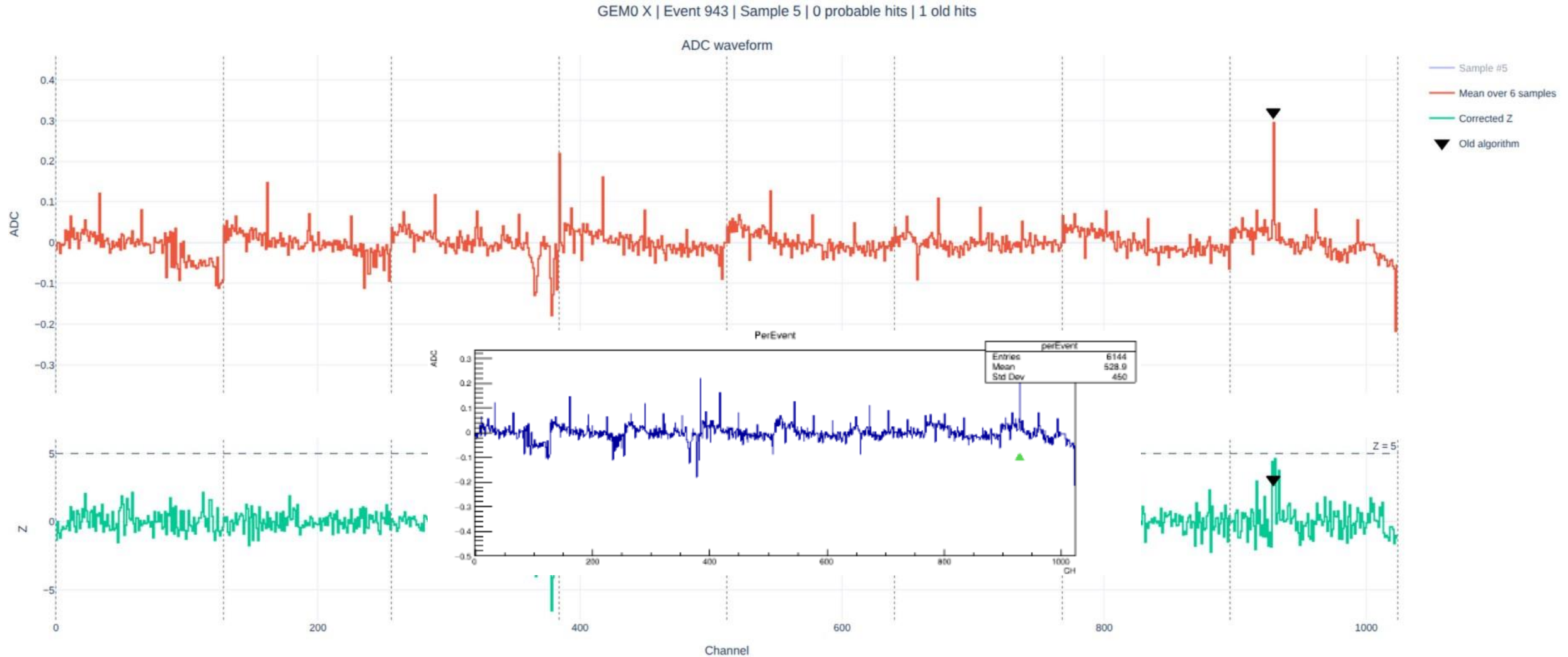
Results: New $\times$ Old $\checkmark$

GEM0-X



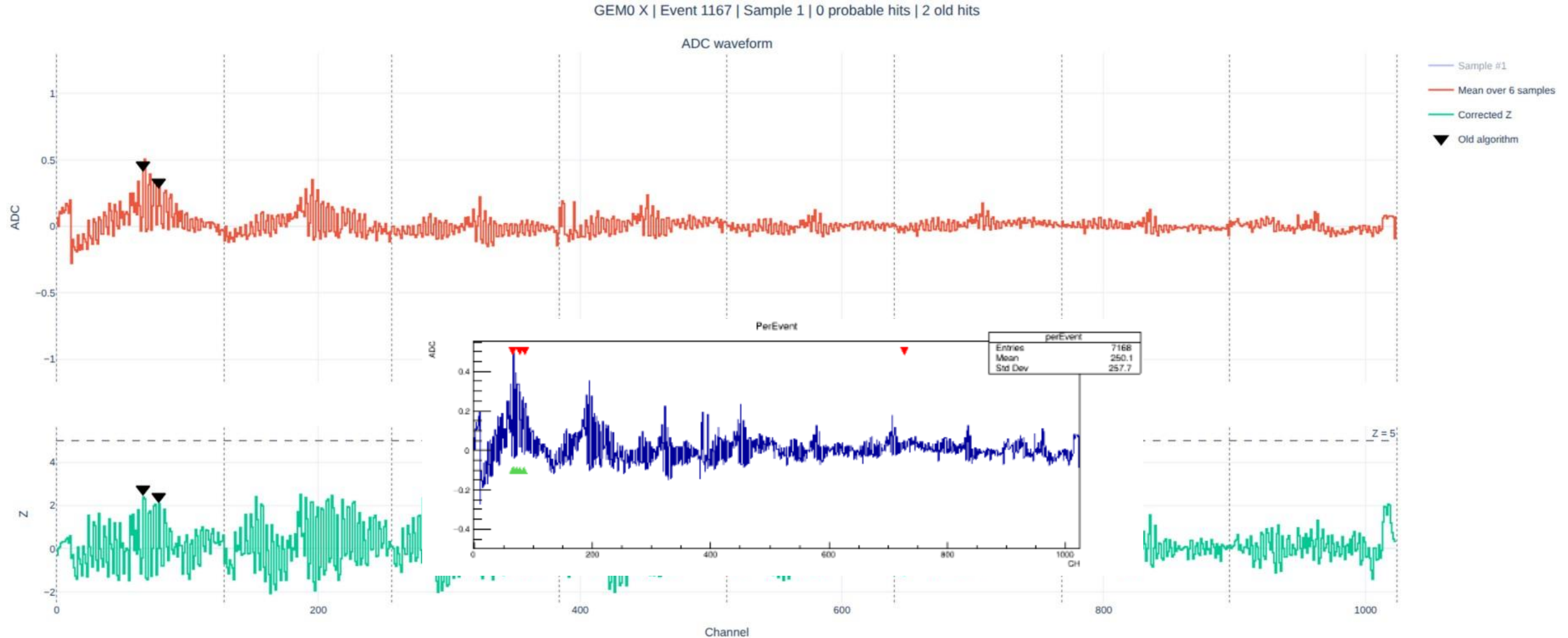
Results: New × Old ✓

GEM0-X

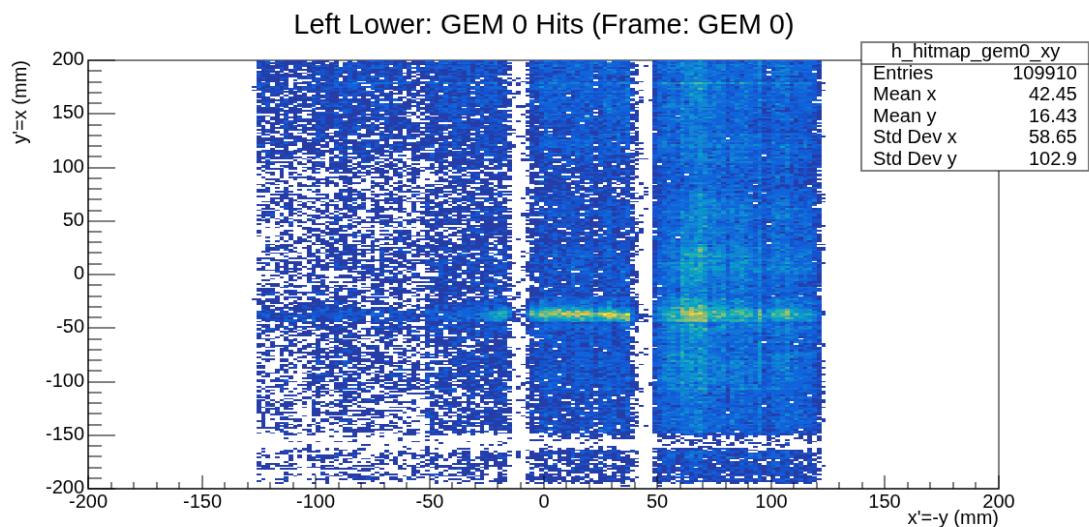
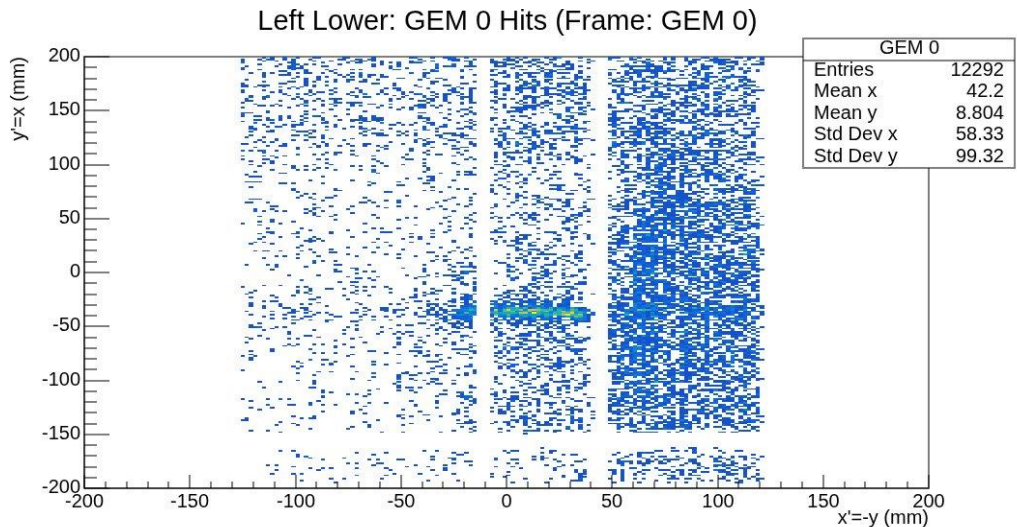
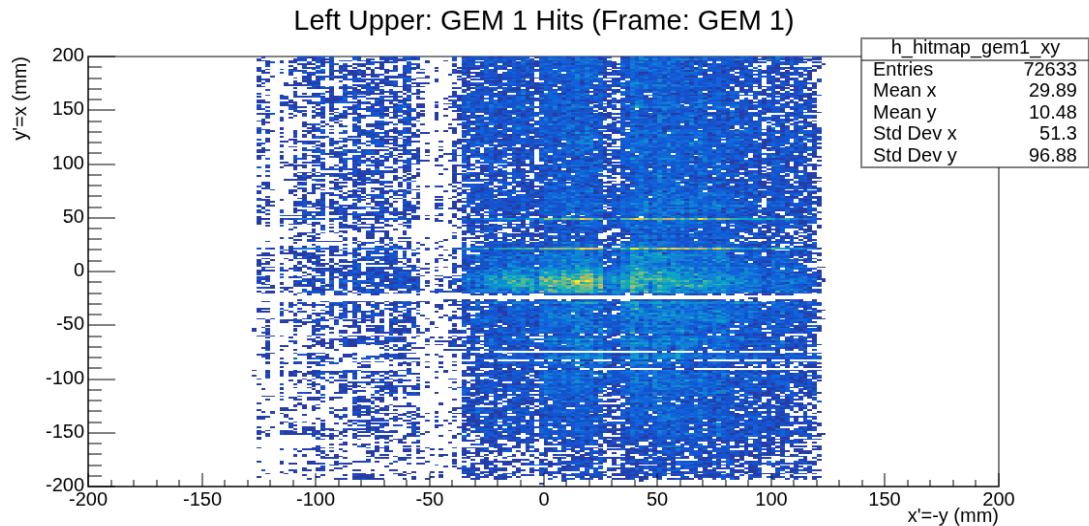
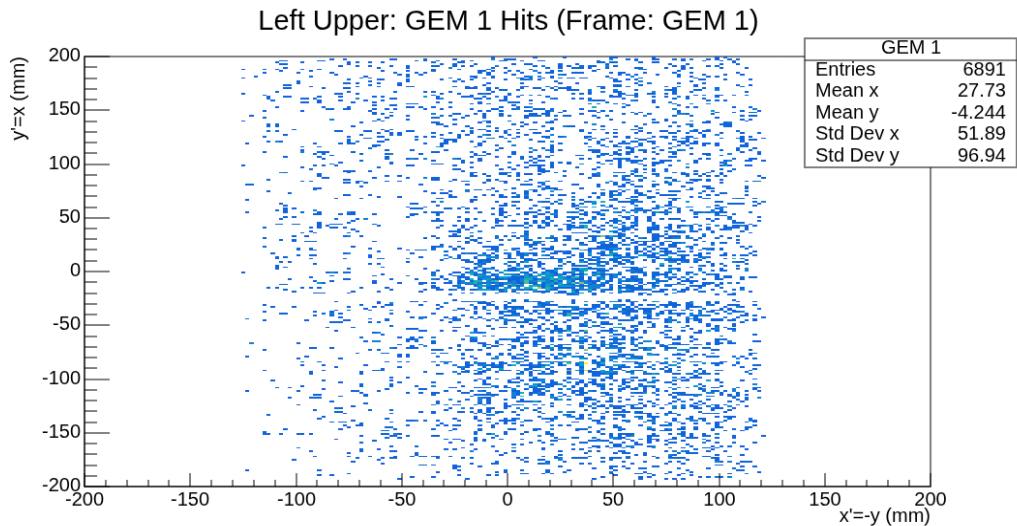


Results: New × Old ✓

GEM0-X



Results: Left Arm



Results: Right Arm

